

Relative Comparison Transport and Projective Spin Fractionalization in Motion–Staticity Geometry

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Abstract A quantum extension of motion–staticity (MS) geometry may contain inter-event comparison structure that is not exhausted by the geometric parallel transport reconstructed from the soldering field. In this paper we develop a constructive model for such an enlarged comparison sector and ask whether purely bosonic comparison degrees of freedom can support emergent spinorial excitations.

Let ω denote the Lorentz connection governing geometric transport and let $\mathcal{A}$ denote an independent Lorentz-covariant quantum comparison connection. Their difference

$$C = \mathcal{A} - \omega$$

is an adjoint-valued one-form rather than a second gauge connection. Using the solder map e , it defines a tangent-space relative-comparison field

$$c_\mu = e^{-1} C_\mu e \in \mathfrak{so}(T_x M, g_x)$$

and a gauge-invariant relative holonomy obtained by comparing the two transports along the same path. We first show that no such nontrivial relative sector can be obtained by transporting the solder map with the same connection on both sides: genuinely new comparison dynamics requires an independent transport structure.

Relative to the timelike tangent of a localized defect or comparison world line, c_μ decomposes into boost and rotational triplets under the compact little group $SO(3)$. If the boost sector is dynamically heavier, the low-energy rotational component provides three bosonic normal modes. The single-quantum subspace transforms as the vector representation,

$$j = 1,$$

without introducing a microscopic spinor. We then couple these continuous relative-comparison modes to a discrete string-occupation variable, so that the vacuum carries no rotational quantum while an occupied comparison-string segment carries one $j = 1$ quantum.

For two adjacent occupied segments, rotational covariance gives the channel decomposition

$$1 \otimes 1 = 0 \oplus 1 \oplus 2,$$

and the most general nearest-neighbor interaction is characterized by three energies E_0, E_1, E_2 . This yields a direct microscopic diagnostic for whether the effective chain lies in the Haldane phase. Ordinary bosonic hopping does not generically satisfy this criterion, so spinorial fractionalization is not a consequence of bosonic comparison quantization alone.

A more specific mechanism is suggested by the representation structure of the comparison bond. The $j = 2$ component of

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{5}$$

is a symmetric-traceless tensor, the same spatial $SO(3)$ representation carried by metric shear. If the comparison vacuum suppresses this shear-like channel with positive stiffness, the leading bond interaction approaches

$$H_{\text{bond}} = \Delta_2 P_{J=2}, \quad \Delta_2 > 0,$$

which is the Affleck–Kennedy–Lieb–Tasaki parent interaction up to normalization and an additive constant. An open comparison string in this phase carries projective spin- $\frac{1}{2}$ boundary modes. Their projective class is detected by the rephasing-invariant rotational comparison quantity

$$\Xi = U_x(\pi)U_y(\pi)U_x(\pi)^{-1}U_y(\pi)^{-1},$$

for which the edge sector has $\Xi = -I$.

The construction is conditional. It does not derive the independent comparison connection, the required compact-sector hierarchy, the string-occupation constraint, the positive $J = 2$ stiffness, or deconfinement from the original MS response axioms. Nor does projective spin- $\frac{1}{2}$ alone imply Fermi statistics. The purpose of the model is instead to reduce the emergence problem to explicit dynamical tests: the quadratic spectrum of the relative comparison field, the three two-body channel energies, the infrared string tension, and the projective boundary invariant. These quantities provide falsifiable criteria for determining whether an enlarged MS comparison dynamics can support deconfined spinorial matter without microscopic spinor variables.

keyword: motion–staticity geometry; soldering form; quantum comparison; relative connection; projective representation; spin fractionalization; Haldane phase; AKLT model; emergent spin; geometric commutant

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1 Introduction

The operational reconstruction of motion–staticity (MS) geometry begins from a generator–displacement response rather than from a prescribed spacetime metric. Under the appropriate composition, continuity, and faithfulness assumptions, the response determines a solder map

$$e : TM \longrightarrow V$$

and hence the Lorentz metric

$$g = e^* \eta.$$

A subsequent quantum analysis shows that an exact local implementation of the same response remains commutative when exact additivity is imposed in both response arguments. A genuinely quantum completion must therefore introduce additional structure, such as conjugate soldering deformations, projective composition, or noncommuting inter-event comparison.

A canonical local completion based on the soldering configuration space

$$\mathcal{E}_x = \text{Hom}(T_x M, V_x)$$

provides one such extension. The operational responses

$$\Theta_{P,\xi}(e) = P(e\xi)$$

form a complete family of linear configuration coordinates, while coherent soldering deformations supply conjugate operators satisfying a Heisenberg–Weyl algebra. Finite rotational comparison provides an independent global quantum structure: the nontrivial projective class of $SO(3)$ can be detected by the rephasing-invariant comparison commutator

$$\Xi = U_x(\pi)U_y(\pi)U_x(\pi)^{-1}U_y(\pi)^{-1}.$$

The value

$$\Xi = -I$$

identifies the half-integer projective rotational class.

These results leave a sharper question than the original problem of “quantizing the geometry.” Canonical noncommutativity does not force a spinorial rotational sector, and the topology of $SO(3)$ permits but does not require the nontrivial projective class. One must therefore explain dynamically how a quantum comparison theory whose microscopic geometric variables are tensorial could produce a sector with

$$\Xi = -I.$$

The present paper addresses this problem constructively.

A first possibility would be to seek the required additional degree of freedom by parallel transporting the solder map around spacetime and comparing the transported soldering with the original one. If, however, the tangent and internal transports are both induced by the same solder map and the same Lorentz connection, the construction is identically trivial. Let

$$U_\gamma^V : V_x \longrightarrow V_y$$

be Lorentz parallel transport generated by ω , and let

$$U_\gamma^T : T_x M \longrightarrow T_y M$$

be the tangent transport induced by e and the same connection. Compatibility of the soldering implies

$$U_\gamma^V e_x = e_y U_\gamma^T.$$

Consequently,

$$e_x^{-1} (U_\gamma^V)^{-1} e_y U_\gamma^T = I.$$

Geometric transport alone therefore cannot generate an independent relative-soldering rotor.

This observation motivates the basic additional structure studied here. We assume that quantum comparison between distinct events may be implemented by a Lorentz-covariant transport

$$C_\gamma$$

that is not required to coincide with the geometric transport U_γ^ω . Locally, let

$$\mathcal{A}$$

be a connection generating the comparison transport and let

$$\omega$$

be the geometric Lorentz connection. Their difference

$$C := \mathcal{A} - \omega \tag{1}$$

transforms homogeneously under the common Lorentz gauge group. It is therefore an adjoint-valued tensorial one-form rather than a second inhomogeneously transforming gauge potential.

Using the solder map, the same information may be represented directly on spacetime tangent space:

$$c_\mu := e^{-1} C_\mu e \in \mathfrak{so}(T_x M, g_x). \tag{2}$$

Equivalently, comparison and geometric transport along the same path define the relative holonomy

$$r_{\gamma,x} := e_x^{-1} \left(U_\gamma^\omega \right)^{-1} C_\gamma e_x, \tag{3}$$

with the obvious endpoint identifications understood. The internal Lorentz gauge transformations cancel from this expression. When both transports are Lorentz isometries,

$$r_{\gamma,x} \in SO^+(T_x M, g_x).$$

Thus a physical relative Lorentz transformation can arise from a mismatch of two comparison structures without introducing an independent second tetrad.

The corresponding relative curvature is controlled by

$$F(\mathcal{A}) - F(\omega) = D_\omega C + C \wedge C. \quad (4)$$

At short distance this quantity measures the failure of comparison transport to coincide with geometric transport. It therefore provides a natural candidate field strength for the enlarged comparison sector.

The noncompactness of the Lorentz group immediately creates a dynamical problem. A naive Lorentz-Yang-Mills action for C does not automatically provide a positive Hamiltonian for all boost and rotational components. The construction pursued here therefore does not assume that the six Lorentz components remain equivalent in the infrared. Instead, a localized comparison defect or world line supplies a timelike tangent

$$u^\mu, \quad u^\mu u_\mu = -1,$$

whose little group is compact:

$$SO(3)_u.$$

Relative to u , the Lorentz algebra decomposes into boost-like and rotation-like triplets. The latter transform as the vector representation of $SO(3)_u$.

The first dynamical question of this paper is whether the rotational triplet can form a low-energy compact sector while the boost triplet remains parametrically heavier. If this occurs, a comparison string with spatial tangent n^μ supports three rotational normal coordinates

$$q^a$$

transforming as

$$\mathbf{3}$$

under $SO(3)_u$. For an approximately isotropic quadratic minimum, their single-quantum states form exactly the spin-one representation:

$$\text{span} \{a_a^\dagger | 0\rangle\} \simeq V_{j=1}.$$

The microscopic bulk degree is therefore an ordinary integer-spin bosonic mode. No spinor has been introduced.

A second problem is occupancy. The existence of a spin-one excitation does not imply that every spatial cell carries one such quantum; the ordinary vacuum is the singlet zero-particle state. We therefore distinguish the continuous relative-comparison field from a discrete comparison-string sector. Let

$$\chi_i \in \{0, 1\}$$

indicate whether a link or segment belongs to an occupied comparison string, and let

$$n_i$$

count the local relative-rotation quanta. A strong binding term of the form

$$H_{\text{bind}} = U \sum_i (n_i - \chi_i)^2 \quad (5)$$

restricts an occupied segment to the one-particle vector sector while leaving an unoccupied segment in the singlet vacuum. This continuous–discrete binding is an additional element of the constructive model; its microscopic origin must ultimately be derived from the comparison dynamics.

Once each occupied segment carries a $j = 1$ mode, the two-segment problem is highly constrained by rotational symmetry:

$$1 \otimes 1 = 0 \oplus 1 \oplus 2. \quad (6)$$

The most general nearest-neighbor $SO(3)$ -invariant interaction is therefore characterized by three channel energies,

$$H_{i,i+1} = E_0 P_{i,i+1}^{(0)} + E_1 P_{i,i+1}^{(1)} + E_2 P_{i,i+1}^{(2)}. \quad (7)$$

This representation provides a direct dynamical diagnostic. The three numbers

$$E_0, \quad E_1, \quad E_2$$

determine whether the effective chain lies in the Haldane phase, in a ferromagnetic or dimerized regime, or at a phase boundary. In particular, ordinary bosonic hopping and Mott-like superexchange do not generically produce the required channel ordering. The desired phase must therefore be selected by comparison-specific dynamics rather than by bosonic statistics alone.

A possible MS-specific mechanism follows from the representation content of the $j = 1$ bond. Under $SO(3)$,

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{5}. \quad (8)$$

The five-dimensional component is symmetric and traceless. The same $\mathbf{5}$ representation appears in the spatial symmetric-traceless metric variation,

$$\delta h_{ab}^{\text{TF}}.$$

Thus the $J = 2$ bond channel is uniquely capable, among the non-scalar bond representations, of carrying the same observer-frame quantum numbers as metric shear.

This observation motivates the comparison-shear hypothesis explored below. If the low-energy comparison sector is required to remain approximately invisible to operational geometry and if shear-like comparison leakage is penalized by a positive response stiffness, then the leading bond term may take the form

$$H_{\text{shear}} = \Delta_2 \sum_i P_{i,i+1}^{(2)}, \quad \Delta_2 > 0. \quad (9)$$

Within the spin-one sector this is the Affleck–Kennedy–Lieb–Tasaki parent interaction, up to normalization and an additive constant. The argument is conditional: representation matching alone does not determine the sign or magnitude of Δ_2 , and integrating out ordinary geometric shear can in some circumstances lower rather than raise the corresponding channel energy. The sign must therefore be established from the renormalized comparison response kernel.

If the effective string lies in the Haldane phase, an open string has projective spin- $\frac{1}{2}$ boundary modes even though every microscopic string segment transforms in an ordinary integer-spin representation. The projective class of an endpoint is measured by the rotational comparison invariant

$$\Xi = U_x(\pi) U_y(\pi) U_x(\pi)^{-1} U_y(\pi)^{-1}. \quad (10)$$

For the nontrivial boundary class,

$$\Xi = -I.$$

The construction therefore supplies a concrete route

bosonic relative comparison $\longrightarrow$ integer-spin string bulk $\longrightarrow$ projective spinorial endpoint.

This result, if dynamically realized, would still not constitute a derivation of a relativistic fermion. A Haldane edge carries half-integer projective spin, but projective spin alone does not determine exchange statistics in a general quantum system. Moreover, the endpoints become particle-like only if the comparison string is deconfined in the infrared. A nonzero string tension

$$E(L) = 2E_{\text{end}} + \sigma L + \dots$$

would confine the two projective endpoints. The emergence of relativistic fermionic matter therefore requires additional continuum conditions, including deconfinement, an appropriate dispersion relation, local relativistic covariance, and ultimately the hypotheses required for a spin–statistics theorem or an independent microscopic statistics mechanism.

The purpose of the present paper is consequently not to assert that these conditions follow automatically. It is to convert the proposed bosonic-to-spinorial mechanism into a sequence of explicit dynamical tests. The construction succeeds only if an underlying comparison theory produces:

1. an independent inter-event comparison transport $\mathcal{A} \neq \omega$;
2. a stable low-energy rotational triplet separated from the boost sector;
3. a mechanism binding one rotational quantum to each occupied comparison string segment;
4. two-segment channel energies E_0, E_1, E_2 lying in the Haldane region, or equivalently a robust positive suppression of the $J = 2$ shear-like channel in the appropriate regime;
5. an infrared phase in which open comparison-string endpoints are deconfined;
6. a continuum sector in which the projective endpoint quantum number $\Xi = -I$ acquires the localization and covariance properties required of physical matter.

Failure of any one of these tests weakens or eliminates the proposed route. The model is therefore intended as a falsifiable constructive framework rather than as a kinematical consequence of the MS response axioms.

The remainder of the paper is organized as follows. Section 2 constructs relative comparison transport and proves the triviality of any relative soldering built from a single common connection. Section 3 analyzes the quadratic relative field, its observer-frame boost–rotation decomposition, and the emergence of a low-energy $j = 1$ sector. Section 4 introduces comparison-string occupation and binds the continuous relative modes to the discrete string support. Section 5 derives the three-channel interaction diagnostic and analyzes the failure of ordinary bosonic superexchange to select the desired phase. Section 6 develops the comparison-shear mechanism, its AKLT limit, and the projective boundary invariant. Section 7 discusses deconfinement, the continuum limit, spin–statistics, and the remaining conditions required for a physical matter sector.

2 Relative Comparison Transport

The construction developed in this paper begins by distinguishing two different notions of transport between events. The first is the geometric Lorentz transport determined by the soldering geometry and its Lorentz connection. The second is an additional quantum comparison transport, whose existence is not implied by the classical soldering reconstruction and is not assumed to coincide with geometric parallel transport.

This distinction is necessary. If the same geometric connection is used on both sides of a comparison, transporting the solder map away from an event and then comparing it with the same soldering structure produces no new relative degree of freedom. We first establish this statement explicitly.

Let

$$e : TM \longrightarrow V$$

be a nondegenerate solder map, and let

$$\omega$$

be a Lorentz connection on V . The pair (e, ω) induces a connection on TM by requiring the soldering map to be covariantly constant. In local form,

$$\nabla^T = e^{-1} \circ \nabla^\omega \circ e, \quad (11)$$

or equivalently,

$$\nabla^\omega e = 0 \quad (12)$$

with respect to the induced tangent connection.

Let

$$\gamma : [0, 1] \longrightarrow M, \quad \gamma(0) = x, \quad \gamma(1) = y,$$

be a piecewise smooth path. Denote the corresponding Lorentz and tangent parallel transports by

$$U_\gamma^\omega : V_x \longrightarrow V_y \quad (13)$$

and

$$U_\gamma^T : T_x M \longrightarrow T_y M. \quad (14)$$

Compatibility of the induced connections implies

$$U_\gamma^\omega e_x = e_y U_\gamma^T. \quad (15)$$

One might attempt to construct a relative soldering transformation by transporting e_y back to x and comparing it with e_x . With the same connection used throughout, however, this construction is identically trivial.

Proposition 2.1 (Single-transport triviality). Let U_γ^ω and U_γ^T be related by Eq. (15). Then

$$e_x^{-1} (U_\gamma^\omega)^{-1} e_y U_\gamma^T = I_{T_x M}. \quad (16)$$

Proof. Equation (15) gives

$$e_y U_\gamma^T = U_\gamma^\omega e_x.$$

Therefore

$$e_x^{-1} (U_\gamma^\omega)^{-1} e_y U_\gamma^T = e_x^{-1} (U_\gamma^\omega)^{-1} U_\gamma^\omega e_x = I_{T_x M}.$$

□

The proposition separates geometric holonomy from the relative-comparison degree sought here. A curved geometric connection may have nontrivial holonomy around closed paths, but comparing that transport with itself does not define an additional relative sector. Any nontrivial relative comparison must compare genuinely different transport structures.

We therefore introduce the principal additional assumption of the present paper.

Independent comparison-transport principle. Inter-event quantum comparison may be implemented by a Lorentz-covariant transport that is not required to coincide with the geometric Lorentz transport determined by ω .

Locally, let

$$\mathcal{A}$$

denote a Lorentz connection generating the independent comparison transport, and write

$$C := \mathcal{A} - \omega. \quad (17)$$

The object C is not another connection. Under a common local Lorentz transformation

$$L : M \longrightarrow SO^+(1, 3),$$

the two connections transform with the same inhomogeneous term. With a standard convention,

$$\omega \longmapsto L\omega L^{-1} - dL L^{-1}, \quad (18)$$

$$\mathcal{A} \longmapsto L\mathcal{A} L^{-1} - dL L^{-1}. \quad (19)$$

Their difference consequently transforms homogeneously:

$$C \longmapsto LCL^{-1}. \quad (20)$$

Thus

$$C \in \Omega^1(M, \text{ad } P_{\text{Lor}}) \quad (21)$$

is an adjoint-valued tensorial one-form.

This distinction is conceptually important. Introducing $\mathcal{A}$ enlarges the comparison dynamics, but the difference C does not introduce a second independent Lorentz gauge redundancy. Both transports are compared under the same geometric Lorentz frame transformation, and their mismatch transforms as an ordinary adjoint-valued field.

The soldering map converts this internal relative field into a tangent-space endomorphism:

$$c_\mu := e^{-1} C_\mu e. \quad (22)$$

Because

$$C_\mu \in \mathfrak{so}(V_x, \eta_x),$$

one has

$$c_\mu \in \mathfrak{so}(T_x M, g_x), \quad g_x = e_x^* \eta_x. \quad (23)$$

Explicitly,

$$g(c_\mu \xi, \zeta) + g(\xi, c_\mu \zeta) = 0. \quad (24)$$

The field c_μ is invariant under the internal Lorentz redundancy when viewed as a geometric endomorphism of $T_x M$. Indeed, under

$$e \longmapsto Le, \quad C_\mu \longmapsto LC_\mu L^{-1},$$

one obtains

$$\begin{aligned} c_\mu &\mapsto e^{-1} L^{-1} L C_\mu L^{-1} L e \\ &= c_\mu. \end{aligned} \quad (25)$$

This statement refers to internal Lorentz gauge transformations; under a change of spacetime coordinates, c_μ transforms in the ordinary tensorial manner.

We now formulate the corresponding finite comparison.

Let

$$\mathcal{U}_\gamma^{\mathcal{A}} : V_x \longrightarrow V_y$$

denote comparison transport generated by $\mathcal{A}$, and let

$$U_\gamma^\omega : V_x \longrightarrow V_y$$

denote geometric transport along the same path. Their relative transport, based at x , is

$$R_{\gamma,x} := \left(U_\gamma^\omega \right)^{-1} \mathcal{U}_\gamma^{\mathcal{A}} : V_x \longrightarrow V_x. \quad (26)$$

Using the solder map, define the tangent-space relative comparison

$$r_{\gamma,x} := e_x^{-1} R_{\gamma,x} e_x : T_x M \longrightarrow T_x M. \quad (27)$$

Both objects compare the two transports along the same geometric path. Neither requires the introduction of a second independent tetrad.

Proposition 2.2 (Gauge covariance and soldered invariance). Under a local Lorentz transformation with endpoint values L_x and L_y ,

$$R_{\gamma,x} \mapsto L_x R_{\gamma,x} L_x^{-1}, \quad (28)$$

while

$$r_{\gamma,x} \mapsto r_{\gamma,x}. \quad (29)$$

Proof. The two transports transform as

$$U_\gamma^\omega \mapsto L_y U_\gamma^\omega L_x^{-1}, \quad (30)$$

$$\mathcal{U}_\gamma^{\mathcal{A}} \mapsto L_y \mathcal{U}_\gamma^{\mathcal{A}} L_x^{-1}. \quad (31)$$

Hence

$$\begin{aligned} R_{\gamma,x} &\mapsto \left(L_y U_\gamma^\omega L_x^{-1} \right)^{-1} \left(L_y \mathcal{U}_\gamma^{\mathcal{A}} L_x^{-1} \right) \\ &= L_x \left(U_\gamma^\omega \right)^{-1} \mathcal{U}_\gamma^{\mathcal{A}} L_x^{-1} \\ &= L_x R_{\gamma,x} L_x^{-1}. \end{aligned} \quad (32)$$

Since

$$e_x \mapsto L_x e_x,$$

one then finds

$$r_{\gamma,x} \mapsto e_x^{-1} L_x^{-1} L_x R_{\gamma,x} L_x^{-1} L_x e_x = r_{\gamma,x}.$$

□

If both ω and $\mathcal{A}$ are Lorentz connections, both parallel transports preserve η . Their relative transport therefore lies in the Lorentz group.

Corollary 2.3 (Relative Lorentz character). If

$$U_\gamma^\omega, \mathcal{U}_\gamma^\mathcal{A} \in SO^+(V, \eta),$$

then

$$R_{\gamma,x} \in SO^+(V_x, \eta_x) \quad (33)$$

and

$$r_{\gamma,x} \in SO^+(T_x M, g_x). \quad (34)$$

Proof. The product of two proper orthochronous Lorentz transformations is again a proper orthochronous Lorentz transformation, so

$$R_{\gamma,x} = (U_\gamma^\omega)^{-1} \mathcal{U}_\gamma^\mathcal{A} \in SO^+(V_x, \eta_x).$$

For $\xi, \zeta \in T_x M$,

$$\begin{aligned} g_x(r_{\gamma,x}\xi, r_{\gamma,x}\zeta) &= \eta_x(R_{\gamma,x}e_x\xi, R_{\gamma,x}e_x\zeta) \\ &= \eta_x(e_x\xi, e_x\zeta) \\ &= g_x(\xi, \zeta). \end{aligned} \quad (35)$$

The orientation and time-orientation properties are inherited by conjugation with e_x . □

The relative comparison is not an independent Wilson line with an ordinary multiplication law. Its composition is twisted by the geometric transport. This is already visible for two composable paths

$$\gamma_1 : x \longrightarrow y, \quad \gamma_2 : y \longrightarrow z.$$

Using

$$U_{\gamma_2 \circ \gamma_1}^\omega = U_{\gamma_2}^\omega U_{\gamma_1}^\omega$$

and the analogous composition law for $\mathcal{U}^\mathcal{A}$, one obtains

$$R_{\gamma_2 \circ \gamma_1, x} = (U_{\gamma_1}^\omega)^{-1} R_{\gamma_2, y} U_{\gamma_1}^\omega R_{\gamma_1, x}. \quad (36)$$

Thus a relative comparison accumulated on the second segment must first be transported back to the initial geometric frame before it can be composed with the first relative comparison.

On tangent space the same relation becomes

$$r_{\gamma_2 \circ \gamma_1, x} = (U_{\gamma_1}^T)^{-1} r_{\gamma_2, y} U_{\gamma_1}^T r_{\gamma_1, x}, \quad (37)$$

where $U_{\gamma_1}^T$ is the tangent transport induced by (e, ω) .

Proposition 2.4 (Twisted comparison composition). The assignment

$$\gamma \longmapsto R_{\gamma,x}$$

does not in general define an independent representation of the path composition law at a fixed fiber. Instead, it obeys Eq. (36). Equivalently, the relative comparison behaves as a cocycle over geometric parallel transport.

This twisted law is important for the physical interpretation of the relative sector. The additional comparison structure is not detached from geometry. It is defined relative to the geometric transport, and its finite composition remembers that background transport through conjugation.

The infinitesimal form of the same statement follows directly from the connection difference. Parametrize a path by s and write

$$\dot{\gamma}^\mu(s)\omega_\mu = \omega_s, \quad \dot{\gamma}^\mu(s)\mathcal{A}_\mu = \mathcal{A}_s, \quad C_s = \mathcal{A}_s - \omega_s.$$

Choose the parallel-transport convention

$$\frac{d}{ds}U^\omega(s) = -\omega_s U^\omega(s), \quad (38)$$

$$\frac{d}{ds}\mathcal{U}^\mathcal{A}(s) = -\mathcal{A}_s \mathcal{U}^\mathcal{A}(s), \quad (39)$$

with both transporters equal to the identity at $s = 0$. For

$$R(s) = U^\omega(s)^{-1} \mathcal{U}^\mathcal{A}(s),$$

one finds

$$\frac{dR}{ds} = -\widetilde{C}_s R, \quad (40)$$

where

$$\widetilde{C}_s := U^\omega(s)^{-1} C_s U^\omega(s). \quad (41)$$

Hence

$$R_{\gamma,x} = \mathcal{P} \exp \left[- \int_0^1 \widetilde{C}_s ds \right]. \quad (42)$$

Equation (42) makes precise the statement that the relative transport is generated by C , but with C continuously referred back to the initial geometric frame.

For an infinitesimal displacement

$$dx^\mu$$

from x , one obtains

$$R_{x+dx,x} = I - C_\mu(x) dx^\mu + O(dx^2), \quad (43)$$

and therefore

$$r_{x+dx,x} = I - c_\mu(x) dx^\mu + O(dx^2). \quad (44)$$

Thus c_μ is the infinitesimal generator of relative comparison.

The two connections also possess different curvatures. With

$$F(\omega) = d\omega + \omega \wedge \omega$$

and

$$F(\mathcal{A}) = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A},$$

substitution of

$$\mathcal{A} = \omega + C$$

gives

$$F(\mathcal{A}) - F(\omega) = D_\omega C + C \wedge C, \quad (45)$$

where

$$D_\omega C = dC + \omega \wedge C + C \wedge \omega. \quad (46)$$

For Lie-algebra-valued one-forms the last expression is equivalently written using the graded adjoint covariant derivative.

Define the relative curvature

$$\mathcal{F}_{\text{rel}} := D_\omega C + C \wedge C. \quad (47)$$

Like C , it transforms homogeneously:

$$\mathcal{F}_{\text{rel}} \mapsto L \mathcal{F}_{\text{rel}} L^{-1}. \quad (48)$$

It therefore measures a genuine mismatch between comparison and geometric transport.

For a sufficiently small closed loop

$$\partial\Sigma$$

based at x , the two holonomies have the standard leading curvature expansions. Their ratio consequently gives

$$R_{\partial\Sigma, x} = I - \mathcal{F}_{\text{rel}\mu\nu}(x) \Sigma^{\mu\nu} + O(\ell^3), \quad (49)$$

up to the common sign convention used for parallel transport and oriented area. On tangent space,

$$r_{\partial\Sigma, x} = I - e_x^{-1} \mathcal{F}_{\text{rel}\mu\nu}(x) e_x \Sigma^{\mu\nu} + O(\ell^3). \quad (50)$$

This establishes a direct analogue of ordinary curvature detection, but with an important difference: the relative loop measures

$$F(\mathcal{A}) - F(\omega)$$

rather than the geometric curvature itself. In the aligned limit

$$\mathcal{A} = \omega,$$

one has

$$C = 0, \quad \mathcal{F}_{\text{rel}} = 0, \quad R_{\gamma, x} = I \quad (51)$$

for every path. The independent comparison sector disappears smoothly.

The preceding results may be summarized as follows.

Theorem 2.5 (Relative-comparison transport). Let e be a nondegenerate solder map, let ω be the geometric Lorentz connection, and let $\mathcal{A}$ be an independent Lorentz-covariant comparison connection transforming under the same local Lorentz group. Then:

1. the difference

$$C = \mathcal{A} - \omega$$

is an adjoint-valued tensorial one-form;

2. its soldered form

$$c = e^{-1} C e$$

is an internally gauge-invariant $\mathfrak{so}(TM, g)$ -valued one-form;

3. comparison and geometric transport along the same path define a relative Lorentz transformation

$$r_{\gamma,x} = e_x^{-1} (U_\gamma^\omega)^{-1} \mathcal{U}_\gamma^{\mathcal{A}} e_x;$$

4. the finite relative transformations obey the twisted composition law Eq. (37);
5. the infinitesimal relative transport is generated by c ;
6. the leading closed-loop mismatch is controlled by

$$\mathcal{F}_{\text{rel}} = D_\omega C + C \wedge C;$$

7. if the comparison transport is identified with the geometric transport, then

$$r_{\gamma,x} = I$$

identically and no independent relative-comparison degree remains.

The theorem identifies the basic dynamical variable of the present paper. The additional comparison sector is not represented by a second independent solder map. It is represented by the relative transport field

$$C = \mathcal{A} - \omega,$$

or equivalently by its soldered tangent-space form c and its finite relative holonomies $r_{\gamma,x}$.

This also sharpens the relation to the quantum-comparison framework developed previously. The exact local response determines the operational soldering geometry, while the canonical quantum completion introduces conjugate local soldering deformations. Neither construction forces an additional inter-event transport. The present paper selects precisely that remaining escape route: an independent comparison transport whose mismatch with geometric parallel transport is dynamically nontrivial.

No dynamics has yet been assigned to C . In particular, the existence of the field does not by itself imply that its six Lorentz-algebra components are healthy propagating modes or that a compact rotational sector is dynamically preferred. The Lorentz algebra is noncompact, and a naive quadratic action based solely on its invariant Killing form need not yield a positive Hamiltonian.

The next section therefore studies the quadratic dynamics of the relative field. A timelike tangent associated with a localized comparison defect will be used to decompose

$$\mathfrak{so}(1, 3)$$

into boost and rotational triplets under a compact $SO(3)$ little group. We shall then determine the conditions under which the rotational triplet forms a stable low-energy sector whose single-quantum states transform as $j = 1$.

3 Compact Relative Modes and the Spin-One Sector

The relative-comparison field constructed in Section 2,

$$c_\mu = e^{-1} C_\mu e, \quad C = \mathcal{A} - \omega,$$

takes values in

$$\mathfrak{so}(T_x M, g_x).$$

At each event it therefore contains six Lorentz-algebra components. The purpose of the present section is to determine under which dynamical conditions a three-component rotational sector can survive as a stable low-energy bosonic mode.

The noncompact Lorentz algebra does not possess a positive-definite invariant quadratic form. Consequently, a naive Yang–Mills-like kinetic term based only on the Lorentz Killing form does not by itself guarantee a healthy positive-energy spectrum for the full relative field. We shall not impose such a microscopic completion. Instead, we formulate the quadratic dynamics relative to a timelike direction supplied by the comparison defect itself.

Let

$$u \in T_x M, \quad g(u, u) = -1,$$

be the future-directed unit tangent to a localized comparison world line, or the timelike tangent to a comparison world sheet. The spatial subspace orthogonal to u is

$$H_u := u^\perp, \quad (52)$$

with positive-definite metric

$$h_{\alpha\beta} = g_{\alpha\beta} + u_\alpha u_\beta. \quad (53)$$

The stabilizer of u inside the proper Lorentz group is

$$SO(H_u) \simeq SO(3)_u.$$

No preferred timelike vector is introduced into the vacuum: u is part of the state or defect under consideration and transforms covariantly under a change of inertial frame.

Write the tangent-space relative field as

$$c_\mu^{\alpha\beta} = -c_\mu^{\beta\alpha}.$$

Relative to u , define the boost-like and rotation-like spatial vectors

$$b_\mu^\alpha := c_\mu^{\alpha\beta} u_\beta, \quad (54)$$

$$\rho_\mu^\alpha := \frac{1}{2} \varepsilon^\alpha_{\beta\gamma\delta} u^\beta c_\mu^{\gamma\delta}, \quad (55)$$

where the spacetime orientation fixes the convention for $\varepsilon_{\alpha\beta\gamma\delta}$.

Both vectors are spatial:

$$u_\alpha b_\mu^\alpha = 0, \quad u_\alpha \rho_\mu^\alpha = 0. \quad (56)$$

Proposition 3.1 (Observer-frame Lorentz decomposition). For fixed unit timelike u , every

$$c_\mu \in \mathfrak{so}(T_x M, g)$$

is uniquely determined by the pair

$$(b_\mu, \rho_\mu) \in H_u \oplus H_u.$$

With the orientation convention of Eq. (55), the inverse decomposition may be written

$$c_\mu^{\alpha\beta} = 2u^{[\alpha} b_\mu^{\beta]} + \varepsilon^{\alpha\beta}_{\gamma\delta} u^\gamma \rho_\mu^\delta. \quad (57)$$

Under the little group $SO(3)_u$, both b_μ and ρ_μ transform as three-dimensional vector representations:

$$\mathfrak{so}(1, 3)|_{SO(3)_u} \simeq \mathbf{3}_{\text{boost}} \oplus \mathbf{3}_{\text{rot}}. \quad (58)$$

Proof. The antisymmetry of $c_\mu^{\alpha\beta}$ implies

$$u_\alpha b_\mu^\alpha = u_\alpha c_\mu^{\alpha\beta} u_\beta = 0.$$

The corresponding orthogonality of ρ_μ follows immediately from the antisymmetry of the Levi–Civita tensor. Substitution of Eq. (57) into Eqs. (54) and (55) recovers b_μ and ρ_μ , respectively, up to the fixed orientation convention. Dimension counting then gives

$$6 = 3 + 3.$$

Elements of the Lorentz stabilizer of u act orthogonally on H_u , so both spatial vectors carry the standard vector representation of $SO(3)_u$. $\square$

The distinction between the two triplets is physical only relative to the state-dependent timelike direction u . In a different inertial frame the decomposition changes covariantly. Hence a dynamical hierarchy between the two sectors need not represent a fundamental violation of Lorentz covariance; it may instead characterize the rest-frame spectrum of a localized defect.

For the string-like sector studied below, let

$$n \in H_u, \quad g(n, n) = 1, \quad g(u, n) = 0,$$

be the unit tangent to a comparison segment. Contracting the one-form index of the relative field with n , define

$$s^\alpha := n^\mu b_\mu^\alpha, \quad (59)$$

$$q^\alpha := n^\mu \rho_\mu^\alpha. \quad (60)$$

Both lie in H_u . The variables s and q will be referred to as the boost and rotational comparison coordinates of the segment.

The origin of their kinetic terms can already be seen in the relative curvature

$$\mathcal{F}_{\text{rel}} = D_\omega C + C \wedge C.$$

On a world sheet with local coordinates

$$(\tau, \sigma),$$

whose timelike and spacelike tangents are aligned locally with u and n , the component

$$\mathcal{F}_{\text{rel } \tau\sigma}$$

contains

$$D_\tau c_\sigma - D_\sigma c_\tau$$

at linear order. In a local comparison frame in which c_τ vanishes at the reference point, and for slowly varying fields along the segment, the rotational component therefore contains

$$D_\tau q$$

to leading order. A positive quadratic response kernel for relative curvature consequently generates a kinetic term for q , and analogously for s .

The most general local quadratic action at lowest derivative order is not fixed by Lorentz covariance alone. Once the defect data u and n are specified, however, invariance under the residual compact group $SO(3)_u$ strongly restricts its form. In a local spatial frame on H_u , write

$$q^a, \quad s^a, \quad a = 1, 2, 3.$$

To quadratic order, an $SO(3)_u$ -invariant world-sheet effective action can be written schematically as

$$\begin{aligned} S_{\text{seg}}^{(2)} = \frac{1}{2} \int d\tau d\sigma & \left[I_\rho (\mathcal{D}_\tau q^a) (\mathcal{D}_\tau q_a) - T_\rho (\mathcal{D}_\sigma q^a) (\mathcal{D}_\sigma q_a) - \kappa_\rho q^a q_a \right. \\ & + I_b (\mathcal{D}_\tau s^a) (\mathcal{D}_\tau s_a) - T_b (\mathcal{D}_\sigma s^a) (\mathcal{D}_\sigma s_a) - \kappa_b s^a s_a \\ & \left. + 2\lambda_0 s^a q_a + 2\lambda_\tau s^a \mathcal{D}_\tau q_a + 2\lambda_\sigma s^a \mathcal{D}_\sigma q_a + \dots \right]. \end{aligned} \quad (61)$$

Here $\mathcal{D}$ denotes the spatial covariant derivative induced on H_u , and the ellipsis denotes higher derivatives and terms involving variations of u or n .

Equation (61) should be regarded as an effective quadratic kernel, not as a fundamental action derived from the MS response axioms. Its coefficients must ultimately be calculated from the microscopic comparison dynamics.

The boost and rotational variables have different parity character. Under a spatial parity transformation in the defect rest frame, s^a is a polar vector whereas q^a is an axial vector. Hence the scalar

$$s \cdot q$$

is parity odd.

Proposition 3.2 (Parity separation of the quadratic sectors). If the defect background and the comparison dynamics preserve spatial parity, then the non-derivative quadratic mixing coefficient satisfies

$$\lambda_0 = 0. \quad (62)$$

Under the corresponding parity assignments, parity-odd derivative mixings are absent as well. The boost and rotational triplets may then be diagonalized independently at quadratic order.

The parity assumption is not required by the kinematics. If parity is broken, the two triplets may mix and the low-energy eigenmodes need not be purely rotational. The projective-spin mechanism considered in this paper therefore has its cleanest realization in the parity-preserving branch.

We now impose the stability conditions

$$I_\rho > 0, \quad T_\rho \geq 0, \quad \kappa_\rho > 0, \quad (63)$$

and

$$I_b > 0, \quad T_b \geq 0, \quad \kappa_b > 0. \quad (64)$$

For a locally homogeneous segment, the two rest-frame gaps are

$$\Delta_\rho = \sqrt{\frac{\kappa_\rho}{I_\rho}}, \quad \Delta_b = \sqrt{\frac{\kappa_b}{I_b}}. \quad (65)$$

More generally, the low-momentum dispersions are

$$\omega_\rho^2(k) = \frac{\kappa_\rho + T_\rho k^2}{I_\rho} + O(k^4), \quad (66)$$

$$\omega_b^2(k) = \frac{\kappa_b + T_b k^2}{I_b} + O(k^4). \quad (67)$$

The existence of a compact low-energy comparison sector requires more than stability. It requires a hierarchy

$$\Delta_b \gg \Delta_\rho. \quad (68)$$

At energies

$$E \ll \Delta_b,$$

the boost triplet may then be integrated out, leaving the rotational triplet as the leading relative-comparison excitation.

We emphasize that Eq. (68) is a dynamical condition. It does not follow from the decomposition

$$\mathfrak{so}(1, 3) = \mathbf{3}_{\text{boost}} \oplus \mathbf{3}_{\text{rot}}$$

alone. A microscopic theory in which the two gaps remain comparable does not reduce to the compact model studied below.

In the low-energy branch satisfying Eq. (68), consider a single localized segment and neglect its spatial gradients. The effective rotational Lagrangian is

$$L_{\text{rot}}^{(2)} = \frac{I_\rho}{2} \dot{q}^a \dot{q}_a - \frac{\kappa_\rho}{2} q^a q_a. \quad (69)$$

The canonical momentum is

$$\pi_a = I_\rho \dot{q}_a, \quad (70)$$

and the Hamiltonian is

$$H_{\text{rot}}^{(2)} = \frac{1}{2I_\rho} \pi_a \pi^a + \frac{\kappa_\rho}{2} q_a q^a. \quad (71)$$

With

$$\omega_\rho = \sqrt{\frac{\kappa_\rho}{I_\rho}}, \quad (72)$$

canonical quantization gives

$$[\widehat{q}^a, \widehat{\pi}_b] = i\hbar \delta_b^a. \quad (73)$$

Define

$$a_a = \sqrt{\frac{I_\rho \omega_\rho}{2\hbar}} \widehat{q}_a + \frac{i}{\sqrt{2I_\rho \hbar \omega_\rho}} \widehat{\pi}_a, \quad (74)$$

$$a_a^\dagger = \sqrt{\frac{I_\rho \omega_\rho}{2\hbar}} \widehat{q}_a - \frac{i}{\sqrt{2I_\rho \hbar \omega_\rho}} \widehat{\pi}_a. \quad (75)$$

They satisfy

$$[a_a, a_b^\dagger] = \delta_{ab}. \quad (76)$$

The Hamiltonian becomes

$$H_{\text{rot}}^{(2)} = \hbar\omega_\rho \left(N + \frac{3}{2} \right), \quad N = a_a^\dagger a_a. \quad (77)$$

The rotational generators acting on this oscillator system are

$$\widehat{J}_a = -i\hbar\epsilon_{abc}a_b^\dagger a_c. \quad (78)$$

They obey

$$[\widehat{J}_a, \widehat{J}_b] = i\hbar\epsilon_{ab}^{c}\widehat{J}_c. \quad (79)$$

Moreover,

$$[\widehat{J}_a, a_b^\dagger] = i\hbar\epsilon_{ab}^{c}a_c^\dagger. \quad (80)$$

Thus the three creation operators transform as the vector representation of $SO(3)_u$.

Theorem 3.3 (Single-quantum spin-one sector). Assume:

1. the relative comparison transport admits a stable aligned vacuum $C = 0$;
2. a defect rest frame u exists;
3. the quadratic comparison kernel preserves $SO(3)_u$;
4. the boost sector is separated by the hierarchy

$$\Delta_b \gg \Delta_\rho;$$

5. the surviving rotational triplet has a stable isotropic quadratic minimum of the form Eq. (71).

Then the one-quantum subspace

$$\mathcal{H}_{N=1} = \text{span} \{a_1^\dagger|0\rangle, a_2^\dagger|0\rangle, a_3^\dagger|0\rangle\} \quad (81)$$

is the irreducible spin-one representation of the defect little group:

$$\mathcal{H}_{N=1} \simeq V_{j=1}. \quad (82)$$

Proof. Equation (80) shows that the three basis states transform as the defining three-dimensional vector representation of $SO(3)_u$. Equivalently, on the one-particle subspace the quadratic Casimir satisfies

$$\widehat{\mathbf{J}}^2 = 2\hbar^2 I. \quad (83)$$

Since

$$j(j+1)\hbar^2 = 2\hbar^2,$$

the representation has

$$j = 1.$$

□

The theorem is the first point in the construction at which an ordinary spin quantum number emerges from the relative-comparison dynamics. Importantly, the result is entirely bosonic. The fundamental relative coordinates q^a are commuting configuration variables before canonical quantization, their creation operators satisfy bosonic oscillator commutation relations, and the one-particle carrier is the ordinary vector representation of $SO(3)$.

No microscopic spinor is present.

The result should also not be confused with the projective spinorial sector that will appear at the boundary of a Haldane phase. At the present stage the rotation group acts linearly:

$$U(2\pi) = I \quad \text{on } \mathcal{H}_{N=1}. \quad (84)$$

The projective invariant introduced in the preceding quantum-comparison work therefore takes its trivial value on a single microscopic comparison quantum:

$$\Xi_{\text{bulk}} = +I. \quad (85)$$

The later appearance of

$$\Xi_{\text{edge}} = -I$$

must consequently be an emergent property of an entangled many-segment state, not a relabeling of the microscopic bulk representation.

This distinction is central to the proposed fractionalization mechanism:

$$\text{microscopic comparison quantum } j = 1, \quad \Xi = +I, \quad (86)$$

while a suitable one-dimensional correlated phase may support

$$\text{boundary comparison mode } j = \frac{1}{2}, \quad \Xi = -I. \quad (87)$$

The isotropic oscillator construction also clarifies what remains to be derived from the microscopic comparison theory. Rotational covariance fixes the degeneracy of the three q^a modes once a compact branch has been selected, but it does not fix the coefficients

$$I_\rho, \quad T_\rho, \quad \kappa_\rho, \quad I_b, \quad T_b, \quad \kappa_b.$$

In particular, neither the positivity conditions Eqs. (63)–(64) nor the hierarchy Eq. (68) follows from the existence of the relative connection alone.

These coefficients may be characterized by response functions of the comparison vacuum. If external sources

$$J_{\rho a}, \quad J_{ba}$$

couple linearly to q^a and s^a , the quadratic one-particle-irreducible effective action has the general momentum-space form

$$\Gamma_{\text{rel}}^{(2)} = \frac{1}{2} \int \frac{d\Omega dk}{(2\pi)^2} \begin{pmatrix} q_a(-p) & s_a(-p) \end{pmatrix} \begin{pmatrix} K_\rho(p) & K_{\rho b}(p) \\ K_{b\rho}(p) & K_b(p) \end{pmatrix} \begin{pmatrix} q^a(p) \\ s^a(p) \end{pmatrix}, \quad (88)$$

where

$$p = (\Omega, k).$$

The factor δ_{ab} has been suppressed. Rotational covariance forces the same kernel for all three spatial components. In a parity-preserving branch,

$$K_{\rho b} = K_{b\rho} = 0 \quad (89)$$

at the parity-even quadratic level.

Near zero momentum one may write

$$K_\rho(\Omega, k) = I_\rho \Omega^2 - T_\rho k^2 - \kappa_\rho + \dots, \quad (90)$$

$$K_b(\Omega, k) = I_b \Omega^2 - T_b k^2 - \kappa_b + \dots. \quad (91)$$

The compact-sector hypothesis is therefore equivalent to a statement about the poles of the comparison two-point functions: the lowest rotational pole must lie parametrically below the lowest boost pole.

This gives a direct dynamical benchmark for any microscopic completion:

$$\frac{\Delta_b}{\Delta_\rho} \gg 1. \quad (92)$$

If the ratio is not large, the three-state spin-one truncation is not a controlled low-energy approximation and the Haldane construction developed later in the paper does not follow.

Even if the hierarchy holds, a further problem remains. The oscillator vacuum

$$|0\rangle$$

is an $SO(3)$ singlet, and nothing in the quadratic relative dynamics forces every spatial segment to carry one quantum. The spin-one sector obtained in Theorem 3.3 is therefore an excitation sector, not the vacuum sector.

The next section introduces a discrete comparison-string occupation variable and a binding interaction that associates one rotational comparison quantum with each occupied string segment. This separates two logically different questions:

what is the lowest nontrivial comparison quantum?

and

where is that quantum required to be present?

The first question is answered here by the $j = 1$ rotational triplet. The second requires additional comparison-string structure.

4 Comparison Strings and Occupancy Binding

Section 3 identified a controlled low-energy branch in which the lowest nontrivial relative-comparison excitation is a bosonic rotational quantum transforming as

$$j = 1$$

under the compact defect little group $SO(3)_u$. This result does not imply that every spatial location carries such a quantum. The oscillator vacuum has

$$N = 0$$

and is an $SO(3)$ singlet. A separate structure is therefore required to specify where the nontrivial relative-comparison sector is present.

We introduce this structure in the form of a discrete comparison-string occupation field. The construction is deliberately minimal. It does not assume a microscopic lattice of spacetime, but it is convenient to formulate the low-energy model on an auxiliary graph or cellular discretization of a spatial slice. Continuum limits will be discussed only after the string dynamics has been specified.

Let

$$\Gamma = (\mathcal{V}, \mathcal{L})$$

be an oriented graph embedded in a spatial hypersurface, with vertices $v \in \mathcal{V}$ and links

$$\ell \in \mathcal{L}.$$

To every link associate a binary occupation variable

$$\chi_\ell \in \{0, 1\}. \quad (93)$$

The interpretation is

$$\chi_\ell = 0$$

for an unoccupied link and

$$\chi_\ell = 1$$

for a segment belonging to a comparison string.

At this stage χ_ℓ is an additional discrete degree of freedom. It is not derived from the local soldering response, from the relative connection $\mathcal{C}$, or from the quadratic oscillator dynamics of Section 3. Its purpose is to distinguish the support of a comparison defect from the continuous field that decorates that support.

On each link, let

$$a_{\ell a}, \quad a_{\ell a}^\dagger, \quad a = 1, 2, 3,$$

denote the low-energy rotational comparison oscillators introduced in Section 3, and define the number operator

$$N_\ell = \sum_{a=1}^3 a_{\ell a}^\dagger a_{\ell a}. \quad (94)$$

The local oscillator Hilbert space decomposes into fixed-number sectors,

$$\mathcal{H}_\ell^{\text{osc}} = \bigoplus_{N=0}^{\infty} \mathcal{H}_{\ell, N}. \quad (95)$$

The vacuum sector is

$$\mathcal{H}_{\ell, 0} \simeq V_{j=0},$$

while Theorem 3.3 gives

$$\mathcal{H}_{\ell, 1} \simeq V_{j=1}.$$

The simplest interaction correlating the discrete string support with the continuous relative-comparison excitation is

$$H_{\text{bind}} = U \sum_{\ell \in \mathcal{L}} (N_\ell - \chi_\ell)^2, \quad U > 0. \quad (96)$$

The coefficient U defines an occupancy-binding scale.

The physical meaning of Eq. (96) is different from that of a chemical potential. It does not by itself favor occupied or unoccupied links. Instead, conditional on the value of χ_ℓ , it energetically correlates the oscillator occupation with the string support.

Proposition 4.1 (Strong-binding occupancy projection). In the limit in which

$$U$$

is parametrically larger than all oscillator, hopping, and bond-interaction scales retained in the low-energy theory, the low-energy Hilbert space on a link satisfies

$$\chi_\ell = 0 \implies N_\ell = 0, \quad (97)$$

$$\chi_\ell = 1 \implies N_\ell = 1. \quad (98)$$

Consequently,

$$\mathcal{H}_\ell^{\chi=0} \simeq V_{j=0}, \quad (99)$$

$$\mathcal{H}_\ell^{\chi=1} \simeq V_{j=1}. \quad (100)$$

Proof. For fixed

$$\chi_\ell \in \{0, 1\},$$

the contribution of link ℓ to Eq. (96) is

$$U(N_\ell - \chi_\ell)^2.$$

Since N_ℓ has nonnegative integer eigenvalues, the unique minimum is

$$N_\ell = \chi_\ell.$$

All sectors with

$$N_\ell \neq \chi_\ell$$

are separated by an energy of order U . In the strong-binding limit they are removed from the low-energy subspace.

For $\chi_\ell = 0$, the surviving oscillator state is the vacuum

$$N_\ell = 0,$$

which is rotationally invariant. For $\chi_\ell = 1$, the surviving one-quantum subspace is the three-dimensional vector representation identified in Theorem 3.3. $\square$

The strong-binding limit therefore realizes a simple local rule:

$$\text{unoccupied segment} \longleftrightarrow j = 0, \quad (101)$$

and

$$\text{occupied comparison segment} \longleftrightarrow j = 1. \quad (102)$$

The spin-one degree of freedom is not the vacuum of the relative oscillator. It is a decoration carried only by the support of the discrete comparison string.

This distinction removes an ambiguity that would arise if one attempted to identify the entire rotor Hilbert space with a spin chain. The ordinary relative-comparison vacuum remains a singlet. A spin-one chain appears only inside a sector in which the discrete comparison support is nontrivial.

For a fixed string configuration χ , let

$$\mathcal{L}_\chi := \{\ell \in \mathcal{L} ; \chi_\ell = 1\} \quad (103)$$

denote its occupied support. In the strong-binding subspace,

$$\mathcal{H}_{\text{low}}(\chi) \simeq \bigotimes_{\ell \in \mathcal{L}_\chi} V_{1,\ell}, \quad (104)$$

up to the singlet factors associated with unoccupied links.

Corollary 4.2 (Spin-one decoration of comparison support). For a fixed occupied comparison string

$$\gamma = (\ell_1, \ell_2, \dots, \ell_N),$$

the strong-binding low-energy Hilbert space is

$$\mathcal{H}_\gamma \simeq V_{1,\ell_1} \otimes V_{1,\ell_2} \otimes \dots \otimes V_{1,\ell_N}. \quad (105)$$

Hence the effective internal dynamics along an occupied string is that of a chain of ordinary integer-spin $j = 1$ degrees of freedom.

The corollary does not determine the interaction among the spin-one factors. It only establishes the local carrier space on which the two-segment Hamiltonian of the next section will act.

The binary support also provides a natural distinction between closed and open comparison strings. Working over $\mathbb{Z}_2$, regard the occupation configuration as a one-chain

$$\chi = \sum_{\ell \in \mathcal{L}} \chi_\ell \ell \in C_1(\Gamma, \mathbb{Z}_2). \quad (106)$$

Its boundary is

$$\partial\chi \in C_0(\Gamma, \mathbb{Z}_2). \quad (107)$$

At a vertex v ,

$$(\partial\chi)_v = \sum_{\ell \ni v} \chi_\ell \pmod{2}. \quad (108)$$

Thus

$$\partial\chi = 0 \quad (109)$$

means that every vertex is incident on an even number of occupied links. In the nonbranching sector this describes a collection of closed loops. Conversely,

$$(\partial\chi)_v = 1 \quad (110)$$

identifies an endpoint of an open comparison string.

The use of $\mathbb{Z}_2$ is not forced by the local relative-comparison field. It is a minimal choice appropriate to an unoriented string-support sector in which only the presence or absence of a comparison segment is retained. More elaborate microscopic completions may require oriented, integer-valued, or higher-form occupation variables.

To allow the comparison support itself to fluctuate, introduce a string Hamiltonian

$$H_{\text{string}}[\chi]. \quad (111)$$

Its precise microscopic form will not be fixed here. A generic effective model may contain a local occupation cost,

$$\mu \sum_{\ell} \chi_{\ell}, \quad (112)$$

together with local reconnection or plaquette terms that change the string configuration while preserving the relevant endpoint sector.

For example, on a lattice admitting elementary plaquettes p , one may schematically write

$$H_{\text{string}} = \mu \sum_{\ell} \chi_{\ell} - K \sum_p \mathcal{R}_p + \dots, \quad (113)$$

where $\mathcal{R}_p$ is a local operator that reconnects the occupied support around the plaquette. The detailed operator algebra depends on the chosen microscopic realization and is not required for the local spin-one construction.

The full low-energy Hamiltonian prior to including spin interactions may therefore be organized as

$$H = H_{\text{string}} + H_{\text{rot}} + H_{\text{bind}} + H_{\text{int}}, \quad (114)$$

where:

- H_{string} governs the support and fluctuation of the discrete comparison strings;
- H_{rot} governs the continuous relative-rotation oscillators;
- H_{bind} correlates oscillator number with string occupation;
- H_{int} contains interactions among neighboring occupied rotational quanta.

Only the last term will determine whether the spin-one string is in a Haldane, ferromagnetic, dimerized, or other correlated phase.

The separation in Eq. (114) is conceptually useful. The existence of comparison strings, the existence of spin-one comparison quanta, and the entanglement phase of those quanta are three distinct dynamical questions. None should be inferred from either of the other two.

In particular,

$$\partial\chi \neq 0$$

does not by itself imply that the endpoints carry half-integer spin. The binary string variable only identifies the endpoint geometry of the support. Its endpoint does not yet possess a projective representation of $SO(3)$.

Proposition 4.3 (No projective endpoint from occupancy alone). The discrete occupation structure

$$\chi_{\ell} \in \mathbb{Z}_2$$

together with the local strong-binding condition

$$N_{\ell} = \chi_{\ell}$$

determines the support of a spin-one decorated string but does not determine the projective rotational class of an open-string endpoint.

Argument. Every occupied microscopic segment carries the ordinary linear $j = 1$ representation and therefore has

$$\Xi = +I.$$

The boundary operator

$$\partial\chi$$

contains only $\mathbb{Z}_2$ support information and no representation data. A value

$$\Xi = -I$$

at an endpoint can therefore arise only from the correlated quantum state of the decorated string, not from string occupancy alone. $\square$

This observation prevents a circular construction. The projective spin- $\frac{1}{2}$ endpoint sought in this paper will not be inserted by hand through the endpoint variable. The microscopic input consists only of integer-spin bosonic degrees of freedom on occupied segments. The nontrivial projective boundary representation must emerge from their many-body entanglement.

The string-support variable and the relative-comparison oscillator also have different transformation properties. The former is taken to be a scalar under the defect-frame rotation group:

$$\chi_\ell \longmapsto \chi_\ell. \quad (115)$$

The latter transforms as

$$V_1.$$

Consequently, the binding interaction

$$(N_\ell - \chi_\ell)^2$$

is rotationally invariant.

If the occupied support forms a nonbranching path

$$\gamma = (\ell_1, \dots, \ell_N),$$

the low-energy theory has a natural one-dimensional ordering inherited from the path. The effective spin operator on segment ℓ_i is

$$\widehat{S}_i^a = -i\hbar\epsilon_{bc}^a a_{ib}^\dagger a_{ic}, \quad (116)$$

restricted to the sector

$$N_i = 1.$$

It satisfies

$$[\widehat{S}_i^a, \widehat{S}_j^b] = i\hbar\delta_{ij}\epsilon^{ab}{}_c \widehat{S}_i^c, \quad (117)$$

and

$$\widehat{\mathbf{S}}_i^2 = 2\hbar^2 I \quad (118)$$

on every occupied segment.

Hence the string-support construction converts the relative-comparison field into a genuine spin-one chain without introducing a microscopic spin-one field as a fundamental variable:

$$C \longrightarrow q^a \longrightarrow a_a^\dagger|0\rangle \longrightarrow V_1 \longrightarrow \chi = 1 \text{ string decoration.} \quad (119)$$

The strong-binding approximation is controlled only when

$$U \gg \Delta_\rho, \quad t_{\text{cmp}}, \quad J_{\text{cmp}}, \quad \dots, \quad (120)$$

where t_{cmp} and J_{cmp} denote characteristic inter-segment transition and interaction scales. At finite U , virtual excursions into

$$N = 0, 2, \dots$$

sectors generate corrections to the effective spin-one Hamiltonian. Such corrections can modify the channel energies derived in the next section and must be included in a microscopic calculation.

This issue is particularly important because ordinary bosonic hopping through virtual double-occupancy states does not generically favor the Haldane phase. The strong-binding projection therefore defines the low-energy carrier space but does not by itself determine a favorable interaction.

The energy of a comparison string also contains information not captured by the spin-one decoration. If the effective string energy at large length is

$$E_\gamma(L) = \epsilon_0 L + E_{\text{spin}}(L) + \dots, \quad (121)$$

with

$$\epsilon_0 > 0,$$

an open-string endpoint will be linearly confined regardless of its eventual projective spin. The condition for deconfined particle-like endpoints will therefore involve the infrared renormalization of the string tension and is logically independent of the local occupancy binding.

For the remainder of the paper we assume only that a regime exists in which:

1. the compact-sector hierarchy of Section 3 is valid;
2. the binding hierarchy Eq. (120) permits projection to

$$N_\ell = \chi_\ell;$$

3. a nonbranching occupied string may be treated locally as an ordered one-dimensional chain of $j = 1$ comparison quanta.

No assumption about the entanglement phase of this chain is made yet.

The next section derives the most general rotationally invariant interaction between two adjacent occupied segments. Because

$$1 \otimes 1 = 0 \oplus 1 \oplus 2,$$

the complete two-segment spectrum is characterized by three channel energies

$$E_0, \quad E_1, \quad E_2.$$

These three quantities provide a direct diagnostic of whether the comparison string can enter the Haldane phase. They also make it possible to test, rather than assume, whether a proposed microscopic comparison interaction supports projective boundary fractionalization.

5 Two-Segment Dynamics and the Three-Channel Test

The strong-binding construction of Section 4 reduces every occupied comparison segment to an ordinary spin-one carrier,

$$\mathcal{H}_i \simeq V_1.$$

The next question is dynamical: which correlated phase is selected when two neighboring occupied segments interact?

Rotational covariance makes this problem unusually economical. For two adjacent occupied segments,

$$V_1 \otimes V_1 = V_0 \oplus V_1 \oplus V_2. \quad (122)$$

Let

$$P_{i,i+1}^{(J)}, \quad J = 0, 1, 2,$$

denote the orthogonal projectors onto the three total-spin sectors. Any nearest-neighbor operator commuting with the diagonal action of $SO(3)_u$ is scalar on each irreducible component. The most general rotationally invariant bond Hamiltonian is therefore

$$H_{i,i+1} = E_0 P_{i,i+1}^{(0)} + E_1 P_{i,i+1}^{(1)} + E_2 P_{i,i+1}^{(2)}, \quad (123)$$

where the three real quantities

$$E_0, \quad E_1, \quad E_2$$

are the renormalized two-segment channel energies.

These quantities are dynamical data. Rotational representation theory determines the existence of the three channels but does not determine their relative ordering.

Proposition 5.1 (Three-channel completeness). On

$$V_1 \otimes V_1,$$

every self-adjoint two-segment Hamiltonian invariant under the diagonal $SO(3)_u$ action has the form Eq. (123). Consequently, up to an overall additive constant, the complete nearest-neighbor rotational dynamics is specified by two independent channel splittings.

Proof. The Clebsch–Gordan decomposition Eq. (122) is multiplicity free. By Schur’s lemma, an operator commuting with the diagonal $SO(3)$ action acts as a scalar on each irreducible summand. Self-adjointness makes these scalars real. $\square$

The channel representation may be converted into the conventional bilinear–biquadratic form. Define dimensionless spin operators

$$\mathbf{s}_i := \frac{\mathbf{S}_i}{\hbar}, \quad (124)$$

so that

$$\mathbf{s}_i^2 = 2$$

on every occupied segment. Introduce

$$x_{i,i+1} := \mathbf{s}_i \cdot \mathbf{s}_{i+1}. \quad (125)$$

Since

$$(\mathbf{s}_i + \mathbf{s}_{i+1})^2 = 4 + 2x_{i,i+1},$$

the eigenvalues of $x_{i,i+1}$ in the three total-spin sectors are

$$\begin{array}{c|ccc} J_{\text{tot}} & 0 & 1 & 2 \\ \hline x & -2 & -1 & 1. \end{array} \quad (126)$$

The corresponding projectors are therefore the interpolation polynomials

$$P^{(0)} = \frac{1}{3}(x^2 - 1), \quad (127)$$

$$P^{(1)} = -\frac{1}{2}(x+2)(x-1), \quad (128)$$

$$P^{(2)} = \frac{1}{6}(x+2)(x+1). \quad (129)$$

In particular,

$$P^{(2)} = \frac{1}{6}[x^2 + 3x + 2]. \quad (130)$$

Substitution into Eq. (123) gives

$$H_{i,i+1} = C + J_1 x_{i,i+1} + J_2 x_{i,i+1}^2, \quad (131)$$

where

$$J_1 = \frac{E_2 - E_1}{2}, \quad (132)$$

$$J_2 = \frac{2E_0 + E_2 - 3E_1}{6}, \quad (133)$$

$$C = \frac{3E_1 + E_2 - E_0}{3}. \quad (134)$$

The additive constant C has no effect on the phase structure.

Equations (132) and (133) provide the central bridge between microscopic comparison dynamics and the standard spin-one phase diagram.

It is conventional to parametrize the nontrivial part of the bond Hamiltonian as

$$H_{i,i+1} - C = J[\cos \theta x_{i,i+1} + \sin \theta x_{i,i+1}^2], \quad J > 0, \quad (135)$$

with

$$J = \sqrt{J_1^2 + J_2^2}, \quad \tan \theta = \frac{J_2}{J_1} \quad (136)$$

whenever $J_1 > 0$.

For the translationally invariant nearest-neighbor spin-one bilinear–biquadratic chain, the standard Haldane phase occupies

$$-\frac{\pi}{4} < \theta < \frac{\pi}{4}. \quad (137)$$

The Heisenberg antiferromagnet lies at

$$\theta = 0,$$

while the Affleck–Kennedy–Lieb–Tasaki point lies inside the same phase.

The criterion Eq. (137) can be written directly in terms of the three comparison-channel energies.

Theorem 5.2 (Three-channel Haldane diagnostic). Assume that the effective occupied comparison string is described, to leading order, by a translation-invariant nearest-neighbor $SO(3)$ -invariant spin-one Hamiltonian of the form Eq. (123). Then the corresponding bilinear–biquadratic coupling lies in the standard Haldane interval

$$-\frac{\pi}{4} < \theta < \frac{\pi}{4}$$

if and only if

$$E_2 > E_1 \tag{138}$$

and

$$3E_1 - 2E_2 < E_0 < E_2. \tag{139}$$

Proof. The Haldane interval in Eq. (137) is equivalent to

$$J_1 > 0$$

and

$$|J_2| < J_1.$$

From Eq. (132),

$$J_1 > 0 \iff E_2 > E_1.$$

Using Eqs. (132) and (133),

$$|J_2| < J_1$$

is

$$|2E_0 + E_2 - 3E_1| < 3(E_2 - E_1).$$

The upper inequality gives

$$E_0 < E_2,$$

while the lower inequality gives

$$E_0 > 3E_1 - 2E_2.$$

Together they give Eq. (139). $\square$

Theorem 5.2 gives a direct microscopic test of the comparison model. No topological invariant or long-chain calculation is required at this stage. Once the renormalized two-segment energies

$$E_0, \quad E_1, \quad E_2$$

are known, one can determine whether the leading spin-one truncation lies inside the Haldane basin.

The two boundaries have simple channel interpretations. The upper boundary

$$E_0 = E_2$$

corresponds to

$$J_2 = J_1,$$

or

$$\theta = \frac{\pi}{4},$$

while

$$E_0 = 3E_1 - 2E_2 \tag{140}$$

corresponds to

$$J_2 = -J_1,$$

or

$$\theta = -\frac{\pi}{4}.$$

A microscopic comparison dynamics that places the three energies far from both relations yields a correspondingly robust low-energy phase.

The diagnostic is invariant under a common shift

$$E_J \mapsto E_J + \Lambda, \quad J = 0, 1, 2,$$

as expected. Only channel splittings have physical significance.

It is useful to identify several special cases.

If

$$E_0 < E_1 < E_2 \tag{141}$$

and the additional inequality in Eq. (139) is satisfied, the bond favors low total spin and has an antiferromagnetic bilinear component.

At the pure bilinear Heisenberg point,

$$J_2 = 0,$$

one has

$$2E_0 + E_2 - 3E_1 = 0. \tag{142}$$

At the AKLT point, only the maximal total-spin channel is penalized:

$$H_{i,i+1}^{\text{AKLT}} = \Delta_2 P_{i,i+1}^{(2)}, \quad \Delta_2 > 0. \tag{143}$$

Hence, up to an additive constant,

$$E_0 = E_1, \quad E_2 - E_0 = \Delta_2 > 0. \tag{144}$$

From Eqs. (132) and (133),

$$J_1 = \frac{\Delta_2}{2}, \quad J_2 = \frac{\Delta_2}{6}, \tag{145}$$

so that

$$\frac{J_2}{J_1} = \frac{1}{3}. \tag{146}$$

Thus the projector Hamiltonian lies strictly inside the Haldane region.

The three-channel formulation also makes it possible to distinguish comparison-specific dynamics from generic bosonic exchange. The latter does not automatically select the required ordering.

Consider the simplest microscopic hopping term for the three bosonic rotational modes,

$$H_t = -t \sum_{\langle ij \rangle} \sum_{a=1}^3 (a_{ia}^\dagger a_{ja} + a_{ja}^\dagger a_{ia}), \tag{147}$$

together with a strong one-particle-per-occupied-segment constraint. A single hop takes the low-energy configuration

$$(1, 1)$$

to a virtual double-occupancy configuration

$$(2, 0) \quad \text{or} \quad (0, 2).$$

If the two bosons occupy the same local orbital, Bose symmetry requires their internal rotational state to be symmetric under exchange. For two spin-one carriers,

$$V_1 \otimes V_1 = V_0^{\text{sym}} \oplus V_1^{\text{antisym}} \oplus V_2^{\text{sym}}. \quad (148)$$

The virtual same-orbital double-occupancy sector therefore contains

$$J = 0 \quad \text{and} \quad J = 2,$$

but not

$$J = 1.$$

Let

$$U_0 > 0, \quad U_2 > 0$$

be the excitation energies of the virtual double-occupancy states in the allowed symmetric channels. Standard second-order degenerate perturbation theory then produces negative energy shifts of the form

$$\Delta E_0^{(2)} = -\alpha_0 \frac{t^2}{U_0}, \quad (149)$$

$$\Delta E_2^{(2)} = -\alpha_2 \frac{t^2}{U_2}, \quad (150)$$

with

$$\alpha_0, \alpha_2 > 0,$$

whereas the antisymmetric channel has no intermediate same-orbital double-occupancy state and therefore receives no shift of this type:

$$\Delta E_1^{(2)} = 0. \quad (151)$$

The precise positive coefficients depend on normalization and microscopic matrix elements; their signs do not.

This yields a useful restricted no-go statement.

Proposition 5.3 (Ordinary-hopping obstruction). Consider a model satisfying the following assumptions:

1. each occupied segment contains one bosonic $j = 1$ comparison quantum in a single local orbital;
2. neighboring segments interact only through rotationally invariant single-particle hopping of the form Eq. (147);
3. the leading interaction in the one-particle-per-site sector is generated at second order through same-orbital double-occupancy states with positive excitation energies U_0, U_2 ;
4. no independent bare bond interaction reverses the resulting channel ordering.

Then the hopping contribution lowers the $J = 2$ channel relative to the $J = 1$ channel:

$$E_2 - E_1 < 0 \quad (152)$$

for nonzero coupling to the $J = 2$ virtual sector. Consequently,

$$J_1 = \frac{E_2 - E_1}{2} < 0, \quad (153)$$

and the pure hopping mechanism does not place the effective chain in the Haldane interval of Theorem 5.2.

Proof. Under the stated assumptions, Eq. (150) gives

$$\Delta E_2^{(2)} < 0,$$

whereas Eq. (151) gives

$$\Delta E_1^{(2)} = 0.$$

If the bare $J = 1$ and $J = 2$ channels are degenerate before hopping, or if no additional interaction reverses the hopping-induced ordering, one obtains

$$E_2 < E_1.$$

Equation (132) then gives

$$J_1 < 0,$$

which violates the necessary Haldane condition $E_2 > E_1$. $\square$

The proposition is intentionally narrow. It does not state that bosonic microscopic models cannot realize the Haldane phase. Additional interactions, multiple orbitals, constrained exchange processes, longer-range terms, or different microscopic Hilbert spaces can change the channel energies. Rather, the result shows that the desired comparison phase is not an automatic consequence of taking the relative modes to be bosonic and imposing one particle per occupied segment.

This negative result is useful for the MS construction. If a Haldane comparison string is realized, the mechanism must contain interaction structure beyond ordinary single-particle bosonic hopping.

A general microscopic comparison interaction may be projected directly onto the three channels. Let

$$V_{\text{cmp}}$$

denote the effective two-segment interaction before projection. Rotational covariance gives

$$E_J = \frac{1}{2J+1} \text{Tr}_{V_J} \left(P^{(J)} V_{\text{cmp}} \right), \quad J = 0, 1, 2, \quad (154)$$

up to a common contribution from the single-segment energies.

Equivalently, if the low-energy relative-comparison theory is formulated through a four-point one-particle-irreducible vertex

$$\Gamma_{ab,cd}^{(4)},$$

rotational covariance implies the decomposition

$$\Gamma^{(4)} = g_0 P^{(0)} + g_1 P^{(1)} + g_2 P^{(2)}. \quad (155)$$

After normalization and matching to the one-particle sector, the renormalized coefficients g_J determine the corresponding channel splittings E_J .

The Haldane test can therefore be implemented directly at the level of the comparison response kernel. In the simplest matching convention one seeks the inequalities

$$g_2 > g_1 \quad (156)$$

and

$$3g_1 - 2g_2 < g_0 < g_2, \quad (157)$$

modulo common shifts and wave-function normalization effects.

This formulation is particularly convenient because it converts the emergence problem into a small set of calculable quantities. A lattice comparison model, perturbative field theory, functional renormalization-group flow, or numerical effective action need not first establish long-distance topological order. It may initially be tested by extracting only the three irreducible two-body kernels.

There is, however, a further structural clue specific to the soldering geometry. The three total-spin channels have different tensor interpretations. If the one-particle vector carrier is written as

$$q^a,$$

then

$$q^a q^b$$

decomposes into:

1. a scalar trace, transforming as **1**;
2. an antisymmetric part, dual to a vector and transforming as **3**;
3. a symmetric-traceless part, transforming as **5**.

The last component is precisely the $J = 2$ sector. It may be represented by the quadrupolar tensor

$$Q^{ab} = q_i^{(a} q_{i+1}^{b)} - \frac{1}{3} h^{ab} q_i \cdot q_{i+1}, \quad (158)$$

with the appropriate operator ordering understood in the quantum theory.

The same five-dimensional $SO(3)$ representation occurs in the symmetric-traceless spatial metric deformation,

$$\delta h_{ab}^{\text{TF}} \in \text{Sym}_0^2(H_u^*) \simeq \mathbf{5}. \quad (159)$$

Thus the $J = 2$ comparison bond is the unique non-scalar two-segment channel with the same observer-frame representation content as metric shear.

Representation matching by itself does not determine an interaction. Nevertheless, it identifies a comparison-specific possibility not present in the ordinary hopping mechanism: the $J = 2$ channel may acquire a distinct stiffness because it is the channel capable of coupling directly to anisotropic geometric response.

The idealized limit in which this channel alone is positively penalized is

$$E_0 = E_1, \quad E_2 > E_0,$$

which, by Eq. (144), is precisely the AKLT projector structure.

The next section develops this possibility. We shall formulate the comparison-shear suppression hypothesis, determine when a positive symmetric-traceless response kernel produces

$$+\Delta_2 P^{(2)},$$

and carefully distinguish such a bare or renormalized comparison stiffness from the effect of merely integrating out an ordinary metric-shear field. The latter can induce the opposite sign. Only after this sign issue has been resolved can the AKLT/Haldane mechanism be interpreted as a genuine comparison-dynamical route to projective spin- $\frac{1}{2}$ endpoints.

6 Comparison Shear Suppression and Projective Endpoints

Section 5 reduced the two-segment dynamics of an occupied comparison string to three rotationally invariant channel energies,

$$E_0, \quad E_1, \quad E_2.$$

The Haldane criterion is therefore completely explicit, but representation theory alone does not determine the ordering of these energies. In particular, ordinary bosonic hopping does not generically produce the required ordering. We now investigate whether the soldering geometry provides a comparison-specific mechanism capable of distinguishing the $J = 2$ channel.

The central observation is representation theoretic. The microscopic comparison quantum on an occupied segment transforms as the vector representation

$$V_1 \simeq \mathbf{3}$$

of the compact defect little group $SO(3)_u$. Hence the two-segment Hilbert space decomposes as

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{5}.$$

The five-dimensional component is the symmetric-traceless tensor representation. The same irreducible representation occurs in the observer-frame decomposition of the spatial metric shear.

We first formulate this correspondence precisely.

Let

$$H_u = u^\perp$$

be the three-dimensional spatial space associated with the timelike defect tangent u , equipped with spatial metric h_{ab} . Complexifying when necessary for quantum amplitudes, the one-particle comparison space may be identified as

$$V_1 \simeq H_u^\mathbb{C}.$$

A two-segment wavefunction may therefore be represented by a tensor

$$\psi^{ab} \in H_u^\mathbb{C} \otimes H_u^\mathbb{C}.$$

It has the unique decomposition

$$\psi^{ab} = \psi_{(0)}^{ab} + \psi_{(1)}^{ab} + \psi_{(2)}^{ab}, \tag{160}$$

where

$$\psi_{(0)}^{ab} = \frac{1}{3} h^{ab} \psi^c{}_c, \quad (161)$$

$$\psi_{(1)}^{ab} = \frac{1}{2} (\psi^{ab} - \psi^{ba}), \quad (162)$$

$$\psi_{(2)}^{ab} = \frac{1}{2} (\psi^{ab} + \psi^{ba}) - \frac{1}{3} h^{ab} \psi^c{}_c. \quad (163)$$

These components transform respectively as

$$J = 0, \quad J = 1, \quad J = 2.$$

The corresponding projector kernels are

$$\left(P^{(0)}\right)_{ab}{}^{cd} = \frac{1}{3} h_{ab} h^{cd}, \quad (164)$$

$$\left(P^{(1)}\right)_{ab}{}^{cd} = \frac{1}{2} (h_a{}^c h_b{}^d - h_a{}^d h_b{}^c), \quad (165)$$

$$\left(P^{(2)}\right)_{ab}{}^{cd} = \frac{1}{2} (h_a{}^c h_b{}^d + h_a{}^d h_b{}^c) - \frac{1}{3} h_{ab} h^{cd}. \quad (166)$$

The last operator is precisely the standard symmetric-traceless projector on a spatial rank-two tensor.

Proposition 6.1 (Bond–shear representation matching). There is a canonical $SO(3)_u$ -equivariant identification

$$V_2 \simeq \text{Sym}_0^2(H_u^C). \quad (167)$$

Under this identification, the total-spin-2 projector on

$$V_1 \otimes V_1$$

is the same algebraic symmetric-traceless projector that isolates the spatial metric-shear representation

$$\delta h_{ab}^{\text{TF}} \in \text{Sym}_0^2(H_u^*).$$

Proof. The tensor product of two vector representations of $SO(3)$ decomposes as

$$H_u \otimes H_u = \Lambda^2 H_u \oplus \text{Sym}^2 H_u.$$

In three dimensions,

$$\Lambda^2 H_u \simeq H_u \simeq V_1,$$

while

$$\text{Sym}^2 H_u = \mathbb{R} h \oplus \text{Sym}_0^2 H_u.$$

The trace component is V_0 , and the five-dimensional symmetric-traceless component is the irreducible V_2 . The explicit projector is Eq. (166). $\square$

The proposition singles out the $J = 2$ channel in a way that is specific to the relation between comparison and geometry. Among the non-scalar two-segment channels, only

$$V_2$$

has the same $SO(3)_u$ representation content as anisotropic spatial metric deformation.

This does not yet determine its energy.

To make the dynamical issue explicit, shift the common two-segment energy so that

$$E_0 = 0$$

and define

$$\delta_1 := E_1 - E_0, \quad \Delta_2 := E_2 - E_0. \quad (168)$$

Then

$$H_{i,i+1} = \delta_1 P_{i,i+1}^{(1)} + \Delta_2 P_{i,i+1}^{(2)}, \quad (169)$$

up to the discarded common constant.

The ideal shear-suppression limit is

$$\delta_1 = 0, \quad \Delta_2 > 0, \quad (170)$$

so that only the symmetric-traceless bond sector is penalized.

It is useful to distinguish three logically different statements:

1. the $J = 2$ bond transforms in the same representation as metric shear;
2. the comparison field may couple to a geometric shear mode;
3. the renormalized $J = 2$ comparison channel may have a positive stiffness relative to the $J = 0, 1$ channels.

Only the first statement follows from representation theory. The third is the condition required for the proposed Haldane mechanism. The second, taken by itself, does not imply the third.

This can be seen from a simple effective-kernel calculation.

Let

$$\phi_A, \quad A = 1, \dots, 5,$$

denote coordinates for the comparison bond in the V_2 channel, and let

$$s_A$$

denote a dynamical geometric shear mode in the corresponding **5** representation. At quadratic order, rotational covariance allows the local effective potential

$$V^{(2)} = \frac{1}{2} \Delta_{\text{cmp}} \phi_A \phi_A + \frac{1}{2} K_h s_A s_A + \lambda s_A \phi_A, \quad (171)$$

with

$$K_h > 0$$

for a stable geometric shear mode.

If the geometric shear is integrated out at Gaussian order, its stationary value is

$$s_A = -\frac{\lambda}{K_h} \phi_A. \quad (172)$$

Substitution into Eq. (171) gives

$$V_{\text{eff}}^{(2)} = \frac{1}{2} \left(\Delta_{\text{cmp}} - \frac{\lambda^2}{K_h} \right) \phi_A \phi_A. \quad (173)$$

Thus ordinary exchange of a stable geometric shear mode lowers the quadratic energy of the comparison $J = 2$ channel.

Proposition 6.2 (Sign of Gaussian shear exchange). Suppose a comparison $J = 2$ mode couples linearly to a stable Gaussian metric-shear mode as in Eq. (171), with

$$K_h > 0.$$

Integrating out the geometric shear produces the correction

$$\delta\Delta_2 = -\frac{\lambda^2}{K_h} \leq 0. \quad (174)$$

Therefore representation matching with metric shear does not by itself generate a positive $J = 2$ penalty.

Proof. The result follows directly from completing the square:

$$\begin{aligned} V^{(2)} = & \frac{K_h}{2} \left(s_A + \frac{\lambda}{K_h} \phi_A \right)^2 \\ & + \frac{1}{2} \left(\Delta_{\text{cmp}} - \frac{\lambda^2}{K_h} \right) \phi_A \phi_A. \end{aligned} \quad (175)$$

□

This proposition rules out an overly simple interpretation of the bond–shear correspondence. A positive gravitational stiffness does not automatically imply a positive comparison $J = 2$ stiffness. If the latter is present, it must arise from an independent bare comparison term, from a more constrained microscopic structure, or from the complete renormalized comparison kernel after all channels have been included.

We therefore formulate the dynamical assumption required below.

Comparison-shear stability hypothesis. After integrating out the microscopic comparison degrees of freedom above the spin-one string scale, the renormalized two-segment interaction has a stable positive symmetric-traceless channel satisfying

$$\Delta_2^{\text{ren}} = E_2^{\text{ren}} - E_0^{\text{ren}} > 0, \quad (176)$$

with the remaining channel splitting sufficiently small, or more generally sufficiently placed, that the inequalities of Theorem 5.2 are satisfied.

The hypothesis is a statement about the full comparison response kernel, not about the metric representation alone. Schematically one may write

$$\Delta_2^{\text{ren}} = \Delta_2^{\text{bare}} + \Pi_2(0), \quad (177)$$

where

$$\Pi_2(0)$$

contains all zero-momentum self-energy and induced-interaction corrections in the $J = 2$ channel. The Gaussian geometric-shear contribution discussed above is one negative term inside $\Pi_2(0)$; other comparison interactions may contribute with either sign.

In the shifted parametrization Eq. (168), the Haldane criterion of Theorem 5.2 becomes particularly simple. Since

$$E_0 = 0, \quad E_1 = \delta_1, \quad E_2 = \Delta_2,$$

one requires

$$\Delta_2 > 0 \tag{178}$$

and

$$\delta_1 < \frac{2}{3}\Delta_2. \tag{179}$$

The ideal shear-suppression point

$$\delta_1 = 0, \quad \Delta_2 > 0$$

therefore lies strictly inside the Haldane region.

At that point,

$$H_{\text{string}}^{\text{spin}} = \Delta_2 \sum_i P_{i,i+1}^{(2)}. \tag{180}$$

Using Eq. (130) from Section 5,

$$P_{i,i+1}^{(2)} = \frac{1}{6} \left[(\mathbf{s}_i \cdot \mathbf{s}_{i+1})^2 + 3\mathbf{s}_i \cdot \mathbf{s}_{i+1} + 2 \right], \tag{181}$$

where

$$\mathbf{s}_i = \mathbf{S}_i / \hbar.$$

Thus Eq. (180) is the AKLT parent Hamiltonian up to an overall normalization and an additive constant [?].

Theorem 6.3 (Comparison-shear AKLT limit). Assume that:

1. every occupied comparison segment is restricted to the microscopic $j = 1$ sector derived in Section 3;
2. the strong-binding projection of Section 4 is valid;
3. the renormalized nearest-neighbor interaction satisfies

$$E_0 = E_1$$

and

$$E_2 - E_0 = \Delta_2 > 0.$$

Then, up to an additive constant, the effective comparison-string Hamiltonian is

$$H_{\text{eff}} = \Delta_2 \sum_i P_{i,i+1}^{(2)}, \tag{182}$$

the spin-one AKLT parent Hamiltonian. Consequently the bulk is in the Haldane phase.

Proof. After subtracting the common value $E_0 = E_1$, the two-segment Hamiltonian contains only

$$\Delta_2 P^{(2)}.$$

Summing over nearest-neighbor occupied segments gives Eq. (182). The standard AKLT construction places this Hamiltonian inside the gapped spin-one Haldane phase [?, ?]. $\square$

The significance of the theorem is not that the AKLT model has been derived from the MS response axioms. It has not. The result is conditional on the renormalized channel ordering. Its significance is that the required interaction has a precise comparison-geometric interpretation: the ideal AKLT parent term is exactly a positive penalty on the symmetric-traceless two-segment channel.

The ground-state structure of Eq. (182) makes the emergence of projective boundary spin explicit. A spin-one carrier may be represented as the symmetric subspace of two auxiliary spin- $\frac{1}{2}$ spaces,

$$V_1 \simeq \text{Sym}^2 V_{1/2}. \quad (183)$$

In the valence-bond representation of the AKLT state, one virtual spin- $\frac{1}{2}$ from each site forms a singlet with a virtual spin- $\frac{1}{2}$ from the neighboring site. Projection of the two virtual spins at each site onto their symmetric subspace reconstructs the physical spin-one carrier.

On a periodic chain all virtual spin- $\frac{1}{2}$ degrees of freedom are paired. On an open chain one virtual spin- $\frac{1}{2}$ remains unpaired at each end. The low-energy edge spaces therefore satisfy

$$\mathcal{H}_{\text{edge,L}} \simeq V_{1/2}, \quad \mathcal{H}_{\text{edge,R}} \simeq V_{1/2}. \quad (184)$$

The microscopic comparison segments nevertheless remain ordinary integer-spin objects:

$$\Xi_{\text{bulk}} = +I. \quad (185)$$

The nontrivial projective representation appears only at the boundary of the entangled many-body state.

To state this result carefully in a gravitational comparison setting, one must distinguish the local Lorentz frame redundancy from the common symmetry used to classify the one-dimensional phase. We therefore assume that along a fixed comparison string the local rotational frames are related to a common reference frame by geometric or relational dressing. Let o be an anchor point on the string and let

$$G_{o \leftarrow i}$$

denote a chosen comparison-compatible transport from segment i to the anchor. A local spin operator may be represented at the anchor as

$$\widetilde{\mathbf{S}}_i = \text{Ad}(G_{o \leftarrow i}) \mathbf{S}_i. \quad (186)$$

Under local internal frame transformations, the dependence on the frame at segment i cancels against that of the transporter. The dressed variables therefore admit a common rotational comparison action at the anchor.

We refer to this assumption as the common rotational-comparison condition. It does not require a preferred global observer in spacetime. It requires only that the one-dimensional defect carry a coherent common $SO(3)$ comparison symmetry after relational dressing. This is the symmetry whose projective action on the edge is measured below.

Let

$$r_x \quad \text{and} \quad r_y$$

be orthogonal π -rotations in this common $SO(3)$. Their ordinary spin-one action on each microscopic segment commutes,

$$U_{\text{bulk}}(r_x)U_{\text{bulk}}(r_y) = U_{\text{bulk}}(r_y)U_{\text{bulk}}(r_x), \quad (187)$$

because the spin-one representation descends to an ordinary representation of $SO(3)$.

At a spin- $\frac{1}{2}$ edge, choose lifts

$$U_{\text{edge}}(r_x) = -i\sigma_x, \quad U_{\text{edge}}(r_y) = -i\sigma_y \quad (188)$$

up to arbitrary overall phases. Their rephasing-invariant group commutator is

$$\begin{aligned} \Xi_{\text{edge}} &:= U_{\text{edge}}(r_x)U_{\text{edge}}(r_y)U_{\text{edge}}(r_x)^{-1}U_{\text{edge}}(r_y)^{-1} \\ &= -I. \end{aligned} \quad (189)$$

Proposition 6.4 (Projective comparison at an AKLT endpoint). Under the common rotational-comparison condition, an open comparison string in the AKLT limit carries an edge sector for which

$$\Xi_{\text{edge}} = -I, \quad (190)$$

whereas every microscopic occupied segment satisfies

$$\Xi_{\text{bulk}} = +I. \quad (191)$$

Hence the nontrivial projective rotational class is generated by boundary fractionalization rather than inserted into the microscopic comparison carrier.

Proof. The microscopic carrier is $j = 1$, so the central element of $SU(2) \rightarrow SO(3)$ acts trivially and

$$\Xi_{\text{bulk}} = +I.$$

The AKLT valence-bond construction leaves an unpaired virtual spin- $\frac{1}{2}$ at each open boundary. On that two-dimensional space, orthogonal π -rotations may be represented as Eq. (188); their commutator is

$$-I.$$

The result is invariant under rephasing of the individual representatives. $\square$

The two endpoints together carry trivial total projective parity:

$$(-1) \times (-1) = +1. \quad (192)$$

Thus a finite open string is consistent with a globally linear microscopic representation even though each endpoint separately transforms projectively.

The exact AKLT point is not required for the projective boundary class. The spin-one Haldane phase is stable under symmetry-preserving perturbations that do not close the bulk gap. Consequently, if the channel energies obey the inequalities of Theorem 5.2 and the common rotational-comparison symmetry is preserved, the nontrivial edge projective class persists throughout the connected Haldane region [?].

Theorem 6.5 (Conditional comparison fractionalization). Assume:

1. the microscopic occupied comparison segments carry only the ordinary integer-spin $j = 1$ representation;
2. their effective nearest-neighbor channel energies satisfy the Haldane conditions of Theorem 5.2;
3. the bulk gap remains open;
4. the common rotational-comparison symmetry is preserved;

5. the string has an open boundary.

Then the boundary sector realizes the nontrivial projective class of $SO(3)$, and the rotational comparison invariant satisfies

$$\Xi_{\text{edge}} = -I. \quad (193)$$

The theorem is a statement about projective spin fractionalization. It does not establish the exchange statistics of the endpoint. In particular,

$$\Xi_{\text{edge}} = -I$$

identifies a half-integer projective rotational sector but does not by itself imply that the endpoint is a relativistic fermion.

There is also an important distinction between the comparison string studied here and the geometric response loop

$$W_\gamma(P) = \exp \left[-\frac{i}{\hbar} \oint_\gamma P_a e^a \right].$$

The latter is constructed from vector and tensor geometric variables and is therefore insensitive to the nontrivial center of the spin cover. Its small-loop behavior probes geometric torsion or related response holonomy. It does not itself create a projective spin- $\frac{1}{2}$ boundary sector.

The comparison string of the present construction plays a different role. Its microscopic segments are decorated by relative-comparison quanta, and its projective endpoint arises from the many-body entanglement phase of those integer-spin degrees of freedom. The two extended structures should therefore not be identified.

The comparison-shear mechanism also admits a direct response-kernel formulation. Let

$$\Gamma_{ab,cd}^{(4)}$$

be the renormalized four-point interaction of neighboring rotational comparison modes after projection to the occupied one-particle sector. Rotational covariance gives

$$\Gamma^{(4)} = g_0 P^{(0)} + g_1 P^{(1)} + g_2 P^{(2)}. \quad (194)$$

Define the shear-channel excess

$$\Delta_{\text{shear}} := g_2 - g_0 \quad (195)$$

and the antisymmetric-channel splitting

$$\delta_{\text{ax}} := g_1 - g_0. \quad (196)$$

The ideal comparison-shear limit is

$$\Delta_{\text{shear}} > 0, \quad \delta_{\text{ax}} = 0. \quad (197)$$

More generally, the Haldane inequalities require

$$\Delta_{\text{shear}} > 0, \quad \delta_{\text{ax}} < \frac{2}{3} \Delta_{\text{shear}}, \quad (198)$$

within the nearest-neighbor truncation and after common shifts and normalizations have been removed.

Equation (198) is the central dynamical target of the present section. It can in principle be tested in a microscopic lattice model, a perturbative comparison theory, a functional effective action, or a numerical calculation without first assuming the existence of spinorial matter.

The comparison-shear hypothesis therefore has a clear failure condition. If the natural renormalized kernel gives

$$g_2 \leq g_1 \quad (199)$$

or violates

$$3g_1 - 2g_2 < g_0 < g_2, \quad (200)$$

then the nearest-neighbor comparison string does not lie in the Haldane region. In that case the projective endpoint mechanism developed here must either be abandoned or supplemented by additional microscopic interactions.

Conversely, a robust region of microscopic parameters satisfying Eq. (198) would establish something stronger than a formal mapping to a known spin chain. It would show that the comparison dynamics itself selects a phase whose boundary carries the nontrivial projective rotational class.

The logical chain established by the present construction is therefore

$$C \longrightarrow j = 1 \text{ relative-comparison quanta} \longrightarrow (E_0, E_1, E_2) \longrightarrow \text{Haldane phase} \longrightarrow \Xi_{\text{edge}} = -I. \quad (201)$$

Only the first arrow is kinematically tied to the relative comparison field. The channel ordering and the Haldane phase remain dynamical.

Even when

$$\Xi_{\text{edge}} = -I$$

is obtained, a final major problem remains. An endpoint of a string with nonzero infrared tension is not an isolated particle. If

$$E(L) = 2M_{\text{edge}} + \sigma_{\text{IR}}L + o(L), \quad (202)$$

with

$$\sigma_{\text{IR}} > 0,$$

the two projective endpoints are linearly confined.

The existence of a spinorial boundary mode must therefore be separated from its deconfinement. The next section studies the infrared string sector, possible loop-condensed regimes, the conditions under which

$$\sigma_{\text{IR}} = 0,$$

and the additional continuum assumptions required before a projective spin- $\frac{1}{2}$ endpoint can be interpreted as relativistic fermionic matter.

7 Deconfinement, Continuum Limit, and Open Problems

The preceding sections establish a conditional microscopic route from an independent relative-comparison transport to an open string carrying a projective spin- $\frac{1}{2}$ boundary sector. This result is not yet a particle construction. An edge mode attached to a string with nonzero infrared tension remains confined, and a half-integer projective representation does not by itself determine relativistic exchange statistics. The purpose of the present section is to state the remaining conditions in a form that can be tested independently.

We begin with confinement.

Consider an open comparison string γ_{xy} whose endpoints are localized near spatial points x and y , with physical separation

$$L = d(x, y).$$

Let

$$E_{\text{open}}(L)$$

denote the lowest energy in the sector containing the two corresponding projective endpoints. At sufficiently large L , define the infrared comparison-string tension by

$$\sigma_{\text{IR}} := \limsup_{L \rightarrow \infty} \frac{E_{\text{open}}(L) - 2M_{\text{end}}}{L}, \quad (203)$$

whenever a finite isolated endpoint energy M_{end} can be defined.

If

$$\sigma_{\text{IR}} > 0,$$

the separation energy grows linearly and the endpoints are confined. A necessary condition for particle-like endpoint excitations is therefore

$$\sigma_{\text{IR}} = 0. \quad (204)$$

This condition is necessary but not by itself sufficient. The large-distance energy must also approach a finite two-endpoint threshold rather than grow sublinearly without bound. A stronger deconfinement criterion is

$$E_{\text{open}}(L) = 2M_{\text{end}} + V_{\text{res}}(L), \quad \lim_{L \rightarrow \infty} V_{\text{res}}(L) = 0, \quad (205)$$

or, more generally, a residual interaction compatible with asymptotically separable endpoint states.

Definition 7.1 (Energetic deconfinement criterion). A comparison-string endpoint sector will be called energetically deconfined if there exists a finite endpoint energy M_{end} such that

$$\lim_{L \rightarrow \infty} E_{\text{open}}(L) = 2M_{\text{end}}, \quad (206)$$

up to interactions that vanish at asymptotic separation.

The distinction between Eq. (204) and Eq. (206) is useful. Vanishing linear tension removes strict confinement but does not exclude logarithmic, Coulombic, or other long-range interactions. The existence of an asymptotic particle interpretation depends on the detailed infrared behavior.

There is also a separate string-breaking issue. If additional microscopic degrees of freedom can screen a string endpoint, the energy of a static open string may saturate at large L because the string breaks into shorter pieces. Such saturation is not by itself evidence that the original projective endpoint has become a deconfined elementary excitation. A proper diagnostic must track the quantum numbers of the endpoint sector, in particular the projective invariant

$$\Xi = -I.$$

Accordingly, let

$$O_{\gamma}(x, y)$$

denote a gauge- and comparison-compatible open-string operator whose endpoint sectors carry the nontrivial projective rotational class. Schematically,

$$O_\gamma(x, y) = \Psi_-(x) \mathcal{S}_\gamma \Psi_-(y),$$

where the notation Ψ_- labels a $\Gamma = -1$ or $\Xi = -I$ endpoint sector and $\mathcal{S}_\gamma$ denotes the required string dressing. No local fundamental spinor field is assumed in this notation.

A deconfined phase should permit the endpoint correlation length and the string-shape dependence to separate in the infrared. In particular, after appropriate renormalization, large deformations of the interior of γ that leave its endpoints fixed should not carry an energy proportional to the added length.

This motivates the following operational criterion.

Criterion 7.2 (Projective endpoint deconfinement). A comparison endpoint with

$$\Xi = -I$$

is a candidate deconfined excitation only if:

1. the asymptotic string tension vanishes;
2. the endpoint energy remains finite;
3. the projective endpoint quantum number survives at arbitrarily large endpoint separation;
4. after infrared dressing, the energy and low-frequency endpoint correlators become insensitive to macroscopic deformations of the interior string path, apart from interactions associated with ordinary long-range fields.

The fourth condition prevents string breaking or arbitrary choice of dressing from being mistaken for deconfinement.

A natural way to seek

$$\sigma_{\text{IR}} = 0$$

is through a fluctuating closed-string phase. The comparison-support Hamiltonian introduced schematically in Section 4 may be organized as

$$H_{\text{string}} = \mu \sum_{\ell} \chi_{\ell} - K \sum_p \mathcal{R}_p + H_{\text{local}} + \cdots, \quad (207)$$

where the first term penalizes occupied length and $\mathcal{R}_p$ implements local reconnection or loop fluctuation. The ratio

$$K/\mu$$

then controls, schematically, a competition between a short-loop regime and a strongly fluctuating loop regime.

No universal critical value of K/μ is implied by the present construction. It depends on the microscopic graph, the operator algebra, the endpoint constraints, and the coupling to the spin-one decoration. The relevant question is instead whether the full decorated-loop model possesses an infrared phase in which arbitrarily large closed comparison strings participate in the ground state without an extensive energy penalty.

A schematic ground state in such a regime may be written

$$|\Omega_{\text{loop}}\rangle = \sum_{\mathcal{L}} \psi(\mathcal{L}) |\mathcal{L}\rangle \otimes |\Phi_{\text{spin}}(\mathcal{L})\rangle, \quad (208)$$

where $\mathcal{L}$ denotes a closed comparison-loop configuration and

$$|\Phi_{\text{spin}}(\mathcal{L})\rangle$$

is the correlated spin-one state decorating its occupied segments. In the phase relevant to the present proposal,

$$|\Phi_{\text{spin}}(\mathcal{L})\rangle$$

lies in the Haldane phase along each sufficiently long loop.

This leads to a useful separation of two gaps. Let

$$\Delta_{\text{H}}$$

denote the bulk Haldane gap of the spin-one decoration, and let

$$\Delta_{\text{loop}}$$

denote the characteristic gap governing local loop fluctuations. Projective endpoint fractionalization is controlled by Δ_{H} , whereas deconfinement depends primarily on the infrared string sector. There is no kinematical reason for these scales to coincide.

A controlled decorated-string regime therefore requires

$$\Delta_{\text{H}} > 0 \tag{209}$$

while simultaneously arranging

$$\sigma_{\text{IR}} = 0. \tag{210}$$

The first protects the projective boundary representation; the second removes linear confinement.

The combination is nontrivial. Closing the Haldane gap destroys the one-dimensional symmetry-protected phase, whereas retaining a positive microscopic string tension confines its edge modes. A successful microscopic comparison dynamics must therefore support an infrared regime in which the spin sector remains gapped while the geometric support of the string becomes tensionless or otherwise deconfined.

Proposition 7.3 (Separation of fractionalization and deconfinement). Within the comparison-string construction, the conditions

$$\Delta_{\text{H}} > 0$$

and

$$\sigma_{\text{IR}} = 0$$

are logically independent. The former protects the projective spin- $\frac{1}{2}$ edge representation of a given long string, whereas the latter determines whether such an edge can be separated from its partner without an energy growing linearly with distance.

The proposition explains why the existence of an AKLT or Haldane description along a fixed string is not sufficient for emergent matter. A conventional open spin-one chain has projective edge states but remains a one-dimensional object. A particle interpretation requires the additional deconfining dynamics of the fluctuating comparison support.

Suppose now that such a deconfined endpoint sector exists. The next question is whether it admits a relativistic continuum description.

Let

$$E(\mathbf{p})$$

denote the lowest endpoint dispersion relation in its rest-frame branch. A necessary infrared condition for an ordinary massive relativistic particle is the existence of constants

$$m_{\text{eff}} > 0, \quad c_{\text{eff}} > 0$$

such that

$$E^2(\mathbf{p}) = m_{\text{eff}}^2 c_{\text{eff}}^4 + c_{\text{eff}}^2 |\mathbf{p}|^2 + O\left(\frac{|\mathbf{p}|^4}{\Lambda_{\text{IR}}^2}\right) \quad (211)$$

for sufficiently small momentum.

The coefficient

$$m_{\text{eff}}$$

is a dynamical endpoint mass. It is not determined by the projective invariant Ξ , by the Haldane gap alone, or by the local MS mass parameter appearing in the classical compact-state construction.

Likewise,

$$c_{\text{eff}}$$

must ultimately agree with the light-cone structure reconstructed by the soldering geometry if the endpoint is to belong to the same relativistic spacetime sector. Thus one requires, after the appropriate normalization,

$$c_{\text{eff}} \longrightarrow c_g, \quad (212)$$

where c_g denotes the characteristic causal speed determined by $g = e^* \eta$.

This requirement is stronger than rotational isotropy. The compact $SO(3)_u$ structure used to construct the spin-one string guarantees only the organization of states in a rest frame. Emergent Lorentz invariance requires the boost sector of the continuum theory to reconstruct the full Poincaré transformation law.

It is therefore useful to distinguish two uses of the boost-like relative modes introduced in Section 3. At the microscopic string scale, the hierarchy

$$\Delta_b \gg \Delta_\rho$$

allows those modes to be integrated out. At the continuum particle level, however, physical Lorentz boosts must still act on the endpoint state through the spacetime symmetry reconstructed from the geometric sector. Suppressing microscopic relative-boost excitations is not the same as eliminating macroscopic Lorentz covariance.

The endpoint Hilbert space must consequently support a representation of the relevant spacetime symmetry. In a massive relativistic sector, the rest-frame little group is

$$SU(2),$$

and the projective endpoint class

$$\Xi = -I$$

selects the half-integer branch. If the lowest such carrier is dynamically selected, the endpoint has

$$j = \frac{1}{2}.$$

At the level of covariant fields, a local spinorial description would use a spin bundle associated with the reconstructed Lorentz geometry. Globally, this requires the corresponding spacetime region to admit a spin structure, or an appropriate generalized spin structure if additional gauge data are present. The existence of the projective boundary mode alone does not solve this global lifting problem.

Suppose that these geometric conditions are satisfied and that the endpoint can be represented in the continuum by a local covariant spinor field

$$\psi.$$

The lowest-derivative effective action then has the familiar schematic form

$$S_{\text{end}} = \int d^4x \sqrt{-g} \left[iZ_\psi \bar{\psi} \gamma^\mu \nabla_\mu \psi - m_{\text{eff}} \bar{\psi} \psi + \mathcal{L}_{\text{int}} + \cdots \right], \quad (213)$$

where this expression is an infrared parametrization, not a microscopic input to the comparison model.

In particular, the spinor ψ in Eq. (213) should not be interpreted as a field that was present in the fundamental relative-comparison action. Its role is analogous to an emergent quasiparticle field used to describe a fractionalized low-energy sector. The microscopic variables remain the relative comparison transport, its bosonic rotational modes, and the comparison-string support.

The transition from

$$\Xi = -I$$

to Fermi statistics requires additional assumptions.

In a local relativistic quantum field theory satisfying the standard hypotheses of a spin–statistics theorem, half-integer spin is associated with fermionic statistics. Accordingly, if the deconfined endpoint sector admits a positive-energy, local, Lorentz-covariant field-theoretic completion with the required causal structure, the projective spin- $\frac{1}{2}$ endpoint must be represented fermionically.

We state this only conditionally.

Conditional Statement 7.4 (Spin–statistics completion). Assume that a deconfined comparison endpoint:

1. carries the nontrivial projective rotational class

$$\Xi = -I;$$

2. belongs to a positive-energy relativistic continuum sector;
3. admits localization by fields satisfying the appropriate causal locality condition;
4. transforms covariantly under the reconstructed spacetime symmetry;
5. satisfies the remaining hypotheses of a relativistic spin–statistics theorem.

Then a lowest spin- $\frac{1}{2}$ endpoint sector is represented by fermionic fields in the continuum theory.

This statement does not constitute a microscopic derivation of Fermi statistics. Rather, it identifies a route by which the bosonic relative-comparison construction can enter the domain of the standard spin–statistics theorem after projective spin fractionalization, deconfinement, and relativistic localization have already been established.

Without these assumptions the endpoint may remain a spin- $\frac{1}{2}$ fractionalized excitation whose exchange statistics must be determined independently.

A microscopic statistics calculation would require additional data not specified by the Haldane construction. Possible diagnostics include the Berry phase associated with adiabatic exchange of two endpoints, the topology of the full multi-defect configuration space, or a field-algebra reconstruction of localized charged sectors. These are distinct from the single-endpoint rotational invariant Ξ .

It is useful to summarize the hierarchy of quantum numbers involved:

$$\begin{aligned}\Xi & : \text{ projective rotational class,} \\ j & : \text{ spin representation,} \\ \vartheta_{\text{exch}} & : \text{ exchange-statistics phase or statistics operator,} \\ m_{\text{eff}} & : \text{ dynamical mass gap.}\end{aligned}\tag{214}$$

The present construction directly targets the first and, under minimality, the second. It does not kinematically determine the third or fourth.

The same caution applies to internal gauge quantum numbers. Nothing in the relative-comparison connection

$$C = \mathcal{A} - \omega$$

or in the Haldane fractionalization mechanism fixes a Standard Model gauge group, a generation structure, or a chiral internal representation. The $SO(3)$ appearing throughout the spin-chain analysis is the compact rest-frame comparison group associated with spacetime rotations, not an internal gauge symmetry.

A future internal gauge sector would have to arise from additional geometry-invisible multiplicity structure, non-Abelian comparison degeneracies, or another independent mechanism. It should not be identified with the Lorentz vector indices of the relative-comparison field.

There is also no prediction here for the numerical value of Newton's constant. The gravitational stiffness encoded in the geometric effective action and the relative-comparison stiffnesses

$$I_\rho, \quad \kappa_\rho, \quad g_0, \quad g_1, \quad g_2$$

may ultimately originate in a common quantum vacuum, but this is a dynamical possibility rather than a result of the present construction.

The main quantitative targets of the model can now be listed without reference to any qualitative interpretation.

A microscopic comparison theory should determine the quadratic kernels

$$K_\rho(\Omega, k), \quad K_b(\Omega, k),\tag{215}$$

and in particular test the compact-sector hierarchy

$$\frac{\Delta_b}{\Delta_\rho} \gg 1.\tag{216}$$

It should determine the occupancy-binding scale

$$U\tag{217}$$

and verify that the one-quantum truncation is controlled.

It should then calculate the three renormalized two-segment channel energies

$$E_0, \quad E_1, \quad E_2,\tag{218}$$

or equivalently

$$g_0, \quad g_1, \quad g_2, \quad (219)$$

and test the Haldane inequalities

$$E_2 > E_1, \quad 3E_1 - 2E_2 < E_0 < E_2. \quad (220)$$

It should separately determine the large-distance string energy and test

$$\sigma_{\text{IR}} = 0, \quad (221)$$

while preserving a finite Haldane bulk gap.

Finally, it should compute the endpoint dispersion and verify whether

$$E^2(\mathbf{p}) - m_{\text{eff}}^2 c_g^4 - c_g^2 |\mathbf{p}|^2 \longrightarrow 0 \quad (222)$$

in the infrared, together with a direct calculation of exchange statistics if a relativistic locality theorem is not independently available.

These conditions define a falsifiable hierarchy:

$$\begin{aligned} \mathcal{A} - \omega &\neq 0, \\ \Delta_b / \Delta_\rho &\gg 1, \\ N_\ell &= \chi_\ell, \\ (E_0, E_1, E_2) &\in \mathcal{R}_{\text{Haldane}}, \\ \bar{\Xi}_{\text{edge}} &= -I, \\ \sigma_{\text{IR}} &= 0, \\ c_{\text{eff}} &= c_g, \end{aligned} \quad (223)$$

where

$$\mathcal{R}_{\text{Haldane}}$$

denotes the channel-energy region specified in Theorem 5.2.

Failure at an early stage invalidates the later interpretation. In particular:

$$\mathcal{A} = \omega \implies C = 0 \implies \text{no independent relative-comparison sector}, \quad (224)$$

$$\Delta_b \sim \Delta_\rho \implies \text{no controlled compact spin-one truncation}, \quad (225)$$

$$(E_0, E_1, E_2) \notin \mathcal{R}_{\text{Haldane}} \implies \text{no Haldane projective endpoint mechanism}, \quad (226)$$

and

$$\sigma_{\text{IR}} > 0 \implies \text{projective endpoints remain confined}. \quad (227)$$

This structure is intentionally stronger than an existence argument. The mechanism can fail at several independently measurable stages.

The conceptual result of the paper may now be stated succinctly. An additional inter-event comparison transport provides an adjoint-valued relative field

$$C = \mathcal{A} - \omega.$$

A compact low-energy rotational branch of this field supplies bosonic spin-one quanta. Binding those quanta to a fluctuating comparison-string support produces a microscopic integer-spin chain. If the renormalized two-segment interaction lies in the Haldane phase, the open string carries a boundary sector with

$$\Xi = -I.$$

If, in addition, the string is deconfined and the endpoint admits a local relativistic continuum completion, the resulting spinorial sector can enter the domain in which the standard spin–statistics connection applies.

The complete conditional chain is therefore

$$\begin{aligned} \text{independent comparison transport} &\longrightarrow C = \mathcal{A} - \omega \\ &\longrightarrow \text{compact rotational triplet} \\ &\longrightarrow j = 1 \text{ bosonic segment quantum} \\ &\longrightarrow \text{spin-one decorated comparison string} \\ &\longrightarrow \text{Haldane entanglement phase} \\ &\longrightarrow \Xi_{\text{edge}} = -I \\ &\longrightarrow \text{deconfined spinorial endpoint} \\ &\longrightarrow \text{relativistic fermionic sector,} \end{aligned} \tag{228}$$

where each arrow after the definition of C contains a dynamical condition rather than a kinematical identity.

The construction therefore stops short of claiming an emergence theorem for ordinary fermionic matter. What it provides is a sequence of explicit mechanisms and obstructions connecting a bosonic relative-comparison field to a projective spin- $\frac{1}{2}$ endpoint. The remaining problem is no longer to guess which representation should be inserted into the theory, but to calculate whether a microscopic comparison dynamics realizes the hierarchy in Eq. (223).

That calculation is the decisive test of the proposal.

A Transport Conventions and Relative-Holonomy Identities

This appendix collects the transport identities used in Section 2 and fixes the associated sign conventions.

Let $P_{\text{Lor}} \rightarrow M$ be the proper orthochronous Lorentz frame bundle. We use matrix notation for the Lorentz connections

$$\omega \quad \text{and} \quad \mathcal{A}.$$

Under a local Lorentz transformation

$$L : M \longrightarrow SO^+(1, 3),$$

our convention is

$$\omega \longmapsto L\omega L^{-1} - dL L^{-1}, \tag{229}$$

and

$$\mathcal{A} \longmapsto L\mathcal{A}L^{-1} - dL L^{-1}. \tag{230}$$

It follows immediately that

$$C := \mathcal{A} - \omega \quad (231)$$

transforms homogeneously:

$$C \mapsto LCL^{-1}. \quad (232)$$

Thus C is a section of

$$T^*M \otimes \text{ad } P_{\text{Lor}}.$$

The curvature convention is

$$F(\omega) = d\omega + \omega \wedge \omega, \quad F(\mathcal{A}) = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A}. \quad (233)$$

Substituting

$$\mathcal{A} = \omega + C$$

gives

$$\begin{aligned} F(\mathcal{A}) &= d\omega + dC + \omega \wedge \omega + \omega \wedge C + C \wedge \omega + C \wedge C \\ &= F(\omega) + D_\omega C + C \wedge C, \end{aligned} \quad (234)$$

where

$$D_\omega C := dC + \omega \wedge C + C \wedge \omega. \quad (235)$$

Therefore

$$F(\mathcal{A}) - F(\omega) = D_\omega C + C \wedge C. \quad (236)$$

Because both curvatures transform homogeneously,

$$F \mapsto LFL^{-1},$$

the relative curvature

$$\mathcal{F}_{\text{rel}} := D_\omega C + C \wedge C \quad (237)$$

satisfies

$$\mathcal{F}_{\text{rel}} \mapsto L\mathcal{F}_{\text{rel}}L^{-1}. \quad (238)$$

Let

$$\gamma : [0, 1] \rightarrow M$$

be a path beginning at x . Along the path write

$$\omega_s = \dot{\gamma}^\mu \omega_\mu, \quad \mathcal{A}_s = \dot{\gamma}^\mu \mathcal{A}_\mu, \quad C_s = \mathcal{A}_s - \omega_s.$$

Our parallel-transport convention is

$$\frac{dU^\omega}{ds} = -\omega_s U^\omega, \quad U^\omega(0) = I, \quad (239)$$

and

$$\frac{dU^\mathcal{A}}{ds} = -\mathcal{A}_s U^\mathcal{A}, \quad U^\mathcal{A}(0) = I. \quad (240)$$

Define

$$R(s) := U^\omega(s)^{-1} U^\mathcal{A}(s). \quad (241)$$

Since

$$\frac{d}{ds} U^\omega(s)^{-1} = U^\omega(s)^{-1} \omega_s,$$

one obtains

$$\begin{aligned} \frac{dR}{ds} &= U^{\omega^{-1}} \omega_s U^{\mathcal{A}} - U^{\omega^{-1}} \mathcal{A}_s U^{\mathcal{A}} \\ &= -U^{\omega^{-1}} C_s U^\omega R. \end{aligned} \quad (242)$$

Defining the interaction-picture relative connection

$$\widetilde{C}_s := U^\omega(s)^{-1} C_s U^\omega(s), \quad (243)$$

gives

$$\frac{dR}{ds} = -\widetilde{C}_s R. \quad (244)$$

Hence

$$R_{\gamma,x} = \mathcal{P} \exp \left[- \int_0^1 \widetilde{C}_s ds \right]. \quad (245)$$

For a sufficiently short path,

$$R_{\gamma,x} = I - C_\mu(x) dx^\mu + O(dx^2). \quad (246)$$

For two composable paths

$$\gamma_1 : x \rightarrow y, \quad \gamma_2 : y \rightarrow z,$$

write

$$R_1 = (U_1^\omega)^{-1} U_1^{\mathcal{A}}, \quad R_2 = (U_2^\omega)^{-1} U_2^{\mathcal{A}}.$$

Then

$$\begin{aligned} R_{\gamma_2 \circ \gamma_1, x} &= (U_2^\omega U_1^\omega)^{-1} U_2^{\mathcal{A}} U_1^{\mathcal{A}} \\ &= (U_1^\omega)^{-1} (U_2^\omega)^{-1} U_2^{\mathcal{A}} U_1^{\mathcal{A}} R_1 \\ &= (U_1^\omega)^{-1} R_2 U_1^\omega R_1. \end{aligned} \quad (247)$$

This is the twisted composition law used in the text.

Let $e_x : T_x M \rightarrow V_x$ be the soldering isomorphism. The tangent-space relative comparison is

$$r_{\gamma,x} = e_x^{-1} R_{\gamma,x} e_x. \quad (248)$$

Under

$$e_x \mapsto L_x e_x$$

and

$$R_{\gamma,x} \mapsto L_x R_{\gamma,x} L_x^{-1},$$

the transformation cancels:

$$r_{\gamma,x} \mapsto r_{\gamma,x}. \quad (249)$$

If

$$U_\gamma^\omega, U_\gamma^{\mathcal{A}} \in SO^+(V, \eta),$$

then

$$R_{\gamma,x} \in SO^+(V_x, \eta_x)$$

and consequently

$$r_{\gamma,x} \in SO^+(T_x M, g_x), \quad g_x = e_x^* \eta_x. \quad (250)$$

For a small closed loop $\partial\Sigma$ based at x , the standard holonomy expansion gives, with the orientation convention used here,

$$U_{\partial\Sigma}^\omega = I - F_{\mu\nu}(\omega)\Sigma^{\mu\nu} + O(\ell^3), \quad (251)$$

$$U_{\partial\Sigma}^{\mathcal{A}} = I - F_{\mu\nu}(\mathcal{A})\Sigma^{\mu\nu} + O(\ell^3). \quad (252)$$

Therefore

$$R_{\partial\Sigma,x} = I - [F(\mathcal{A}) - F(\omega)]_{\mu\nu} \Sigma^{\mu\nu} + O(\ell^3), \quad (253)$$

or

$$R_{\partial\Sigma,x} = I - \mathcal{F}_{\text{rel}\mu\nu} \Sigma^{\mu\nu} + O(\ell^3). \quad (254)$$

Changing the sign convention for parallel transport reverses the common sign in these two small-loop formulas but leaves all relative statements unchanged.

B Lorentz-Algebra Decomposition Relative to a Timelike Defect

We use metric signature

$$(-, +, +, +).$$

Let

$$u^\alpha u_\alpha = -1$$

be a future-directed unit timelike vector and define the spatial projector

$$h^\alpha_\beta = \delta^\alpha_\beta + u^\alpha u_\beta. \quad (255)$$

Its image is

$$H_u = u^\perp.$$

Let

$$c^{\alpha\beta} = -c^{\beta\alpha}$$

be an arbitrary Lorentz-algebra element. Define its boost component by

$$b^\alpha := c^{\alpha\beta} u_\beta. \quad (256)$$

Antisymmetry immediately gives

$$u_\alpha b^\alpha = 0.$$

To avoid orientation-dependent sign ambiguities in the rotational component, define the induced spatial Levi–Civita tensor

$$\varepsilon_{u\alpha\beta\gamma} := u^\delta \varepsilon_{\delta\alpha\beta\gamma}, \quad (257)$$

with its sign fixed once the spacetime orientation and the future direction of u are chosen.

The purely spatial part of c is

$$c_{\perp}^{\alpha\beta} := h^{\alpha}_{\gamma} h^{\beta}_{\delta} c^{\gamma\delta}. \quad (258)$$

Because H_u is three-dimensional, spatial Hodge duality gives a unique vector $\rho^{\alpha} \in H_u$ such that

$$c_{\perp}^{\alpha\beta} = \varepsilon_u^{\alpha\beta}{}_{\gamma} \rho^{\gamma}. \quad (259)$$

Equivalently,

$$\rho^{\alpha} = \frac{1}{2} \varepsilon_u^{\alpha}{}_{\beta\gamma} c_{\perp}^{\beta\gamma}. \quad (260)$$

The full Lorentz-algebra element is then reconstructed as

$$c^{\alpha\beta} = 2u^{[\alpha} b^{\beta]} + \varepsilon_u^{\alpha\beta}{}_{\gamma} \rho^{\gamma}. \quad (261)$$

The first term contains the three boost generators and the second the three spatial rotations.

The map

$$\mathfrak{so}(1, 3) \longrightarrow H_u \oplus H_u, \quad c \longmapsto (b, \rho), \quad (262)$$

is therefore a linear isomorphism.

Under the stabilizer

$$SO(3)_u = \{\Lambda \in SO^+(1, 3) ; \Lambda u = u\},$$

both b and ρ transform as spatial vectors. Hence

$$\mathfrak{so}(1, 3)|_{SO(3)_u} \simeq \mathbf{3} \oplus \mathbf{3}. \quad (263)$$

Under spatial parity in the u -rest frame, b is polar while ρ is axial. Consequently,

$$b \cdot \rho \quad (264)$$

is parity odd. Thus a parity-preserving local quadratic potential cannot contain a nonderivative $b \cdot \rho$ mixing term.

For a string segment with unit spatial tangent

$$n \in H_u, \quad n^{\alpha} n_{\alpha} = 1,$$

the one-form index of the relative comparison field may be contracted with n . Writing

$$c_{\mu}^{\alpha\beta}$$

for the full field, define

$$s^{\alpha} = n^{\mu} b_{\mu}^{\alpha}, \quad q^{\alpha} = n^{\mu} \rho_{\mu}^{\alpha}. \quad (265)$$

For fixed defect data (u, n) , both are three-component variables on H_u . The subsequent spin-one construction uses the rotational coordinate q .

C Quadratic Relative Modes and the Spin-One Oscillator

Let

$$q^a, \quad s^a, \quad a = 1, 2, 3,$$

denote the rotational and boost coordinates associated with one local comparison segment in its rest frame.

Rotational covariance implies that the quadratic one-particle-irreducible kernel is proportional to δ_{ab} in the internal spatial indices. In momentum space,

$$\Gamma^{(2)} = \frac{1}{2} \int_p \begin{pmatrix} q_a(-p) & s_a(-p) \end{pmatrix} \begin{pmatrix} K_\rho(p) & K_{\rho b}(p) \\ K_{b\rho}(p) & K_b(p) \end{pmatrix} \begin{pmatrix} q^a(p) \\ s^a(p) \end{pmatrix}. \quad (266)$$

Here

$$\int_p := \int \frac{d\Omega dk}{(2\pi)^2}$$

for the effective world-sheet theory.

In a parity-preserving branch the nonderivative q - s mixing vanishes. If the remaining derivative mixing is absent by the microscopic discrete symmetries or removed by diagonalization of the quadratic kernel, one may write

$$K_\rho(\Omega, k) = I_\rho \Omega^2 - T_\rho k^2 - \kappa_\rho + O(p^4), \quad (267)$$

$$K_b(\Omega, k) = I_b \Omega^2 - T_b k^2 - \kappa_b + O(p^4). \quad (268)$$

Stability of the two quadratic branches requires

$$I_\rho > 0, \quad T_\rho \geq 0, \quad \kappa_\rho > 0, \quad (269)$$

and

$$I_b > 0, \quad T_b \geq 0, \quad \kappa_b > 0. \quad (270)$$

Their rest-frame gaps are

$$\Delta_\rho = \sqrt{\frac{\kappa_\rho}{I_\rho}}, \quad \Delta_b = \sqrt{\frac{\kappa_b}{I_b}}. \quad (271)$$

The compact-sector approximation requires

$$\Delta_b \gg \Delta_\rho. \quad (272)$$

At energies well below Δ_b , the boost variables may be integrated out and the leading local rotational Hamiltonian is

$$H_{\text{rot}} = \frac{1}{2I_\rho} \pi_a \pi_a + \frac{I_\rho \omega_\rho^2}{2} q_a q_a, \quad \omega_\rho = \sqrt{\frac{\kappa_\rho}{I_\rho}}. \quad (273)$$

Canonical quantization gives

$$[q_a, \pi_b] = i\hbar \delta_{ab}. \quad (274)$$

Define

$$a_a = \sqrt{\frac{I_\rho \omega_\rho}{2\hbar}} q_a + \frac{i}{\sqrt{2I_\rho \hbar \omega_\rho}} \pi_a, \quad (275)$$

$$a_a^\dagger = \sqrt{\frac{I_\rho \omega_\rho}{2\hbar}} q_a - \frac{i}{\sqrt{2I_\rho \hbar \omega_\rho}} \pi_a. \quad (276)$$

Then

$$[a_a, a_b^\dagger] = \delta_{ab}$$

and

$$H_{\text{rot}} = \hbar\omega_\rho \left(N + \frac{3}{2} \right), \quad N = a_a^\dagger a_a. \quad (277)$$

The $SO(3)$ generators are

$$J_a = -i\hbar\epsilon_{abc}a_b^\dagger a_c. \quad (278)$$

Using the oscillator algebra gives

$$[J_a, J_b] = i\hbar\epsilon_{ab}^{c} J_c, \quad (279)$$

and

$$[J_a, a_b^\dagger] = i\hbar\epsilon_{ab}^{c} a_c^\dagger. \quad (280)$$

Therefore

$$a_a^\dagger$$

transforms as the vector representation.

On the one-particle subspace

$$\mathcal{H}_{N=1} = \text{span}\{a_1^\dagger|0\rangle, a_2^\dagger|0\rangle, a_3^\dagger|0\rangle\}, \quad (281)$$

one may evaluate the quadratic Casimir explicitly. Acting on

$$|d\rangle := a_d^\dagger|0\rangle,$$

one has

$$J_a|d\rangle = i\hbar\epsilon_{ad}^{c}|c\rangle. \quad (282)$$

Therefore

$$\begin{aligned} \mathbf{J}^2|d\rangle &= -\hbar^2\epsilon_{ad}^{c}\epsilon_{ac}^{e}|e\rangle \\ &= 2\hbar^2|d\rangle, \end{aligned} \quad (283)$$

where the three-dimensional epsilon identity has been used. Hence

$$\mathbf{J}^2 = 2\hbar^2 I \quad \text{on } \mathcal{H}_{N=1}. \quad (284)$$

Since

$$j(j+1)\hbar^2 = 2\hbar^2,$$

one obtains

$$j = 1. \quad (285)$$

This calculation also makes clear that the microscopic one-particle representation is linear under $SO(3)$:

$$U(2\pi) = I.$$

Its projective invariant is therefore

$$\Xi_{\text{bulk}} = +I.$$

D Strong Occupancy Binding and the Low-Energy String Hilbert Space

Let

$$\chi_\ell \in \{0, 1\}$$

denote the binary comparison-string occupation of a link ℓ , and let

$$N_\ell = a_{\ell a}^\dagger a_{\ell a}.$$

Consider

$$H_0 = U \sum_\ell (N_\ell - \chi_\ell)^2, \quad U > 0. \quad (286)$$

For fixed χ_ℓ , the spectrum of the local binding term is

$$U(N - \chi_\ell)^2, \quad N = 0, 1, 2, \dots$$

The unique local minimum is

$$N = \chi_\ell.$$

Therefore the exact ground subspace of H_0 for a fixed occupation configuration is

$$\mathcal{H}_{\text{low}}(\chi) = \bigotimes_{\chi_\ell=0} \mathcal{H}_{\ell, N=0} \otimes \bigotimes_{\chi_\ell=1} \mathcal{H}_{\ell, N=1}. \quad (287)$$

Since

$$\mathcal{H}_{N=0} \simeq V_0$$

and

$$\mathcal{H}_{N=1} \simeq V_1,$$

one obtains

$$\mathcal{H}_{\text{low}}(\chi) \simeq \bigotimes_{\chi_\ell=1} V_{1,\ell}, \quad (288)$$

up to the one-dimensional vacuum factors on unoccupied links.

Let

$$V$$

denote all remaining weak terms:

$$H = H_0 + V.$$

Introduce the projector

$$P$$

onto $\mathcal{H}_{\text{low}}(\chi)$ and

$$Q = I - P.$$

When

$$\|V\| \ll U,$$

a Schrieffer–Wolff or degenerate perturbative expansion gives

$$H_{\text{eff}} = PHP - PVQ \frac{1}{QH_0Q - E_0} QVP + O\left(\frac{\|V\|^3}{U^2}\right). \quad (289)$$

Thus the projection

$$N_\ell = \chi_\ell$$

is controlled only when all retained inter-segment scales are small compared with U .

The leading corrections scale as

$$\delta H_{\text{eff}} = O\left(\frac{V^2}{U}\right). \quad (290)$$

These terms can modify the three channel energies and therefore cannot be ignored when a quantitative microscopic model is constructed.

Working over $\mathbb{Z}_2$, the occupation configuration defines a one-chain

$$\chi = \sum_{\ell} \chi_{\ell} \ell \in C_1(\Gamma, \mathbb{Z}_2). \quad (291)$$

Its boundary is

$$\partial\chi = \sum_v \left(\sum_{\ell \ni v} \chi_{\ell} \right) v \pmod{2}. \quad (292)$$

Hence a vertex belongs to

$$\partial\chi$$

precisely when an odd number of occupied links meet there.

The support variable carries no spin representation. Therefore

$$\partial\chi \neq 0$$

does not imply

$$\Xi = -I.$$

The projective endpoint quantum number must arise from the entangled state of the V_1 decorations.

E Spin-One Projectors and the Three-Channel Haldane Criterion

Let

$$\mathbf{s}_i = \frac{\mathbf{S}_i}{\hbar}, \quad \mathbf{s}_i^2 = 2.$$

Define

$$x := \mathbf{s}_i \cdot \mathbf{s}_{i+1}. \quad (293)$$

For total spin J ,

$$(\mathbf{s}_i + \mathbf{s}_{i+1})^2 = J(J+1).$$

Since

$$(\mathbf{s}_i + \mathbf{s}_{i+1})^2 = 4 + 2x,$$

one obtains

$$x_J = \frac{1}{2} [J(J+1) - 4]. \quad (294)$$

Therefore

$$x_0 = -2, \quad x_1 = -1, \quad x_2 = 1. \quad (295)$$

Because the spectrum of x consists of three distinct values, the projectors are obtained by Lagrange interpolation:

$$\begin{aligned} P^{(0)} &= \frac{(x+1)(x-1)}{(-2+1)(-2-1)} \\ &= \frac{1}{3}(x^2 - 1), \end{aligned} \quad (296)$$

$$\begin{aligned} P^{(1)} &= \frac{(x+2)(x-1)}{(-1+2)(-1-1)} \\ &= -\frac{1}{2}(x+2)(x-1), \end{aligned} \quad (297)$$

$$\begin{aligned} P^{(2)} &= \frac{(x+2)(x+1)}{(1+2)(1+1)} \\ &= \frac{1}{6}(x+2)(x+1). \end{aligned} \quad (298)$$

They satisfy

$$P^{(0)} + P^{(1)} + P^{(2)} = I, \quad P^{(J)}P^{(K)} = \delta_{JK}P^{(J)}. \quad (299)$$

Let

$$H_{\text{bond}} = E_0P^{(0)} + E_1P^{(1)} + E_2P^{(2)}. \quad (300)$$

Using Eqs. (296)–(298),

$$H_{\text{bond}} = C + J_1x + J_2x^2, \quad (301)$$

where

$$J_1 = \frac{E_2 - E_1}{2}, \quad (302)$$

$$J_2 = \frac{2E_0 + E_2 - 3E_1}{6}, \quad (303)$$

$$C = \frac{3E_1 + E_2 - E_0}{3}. \quad (304)$$

Conversely,

$$E_0 = C - 2J_1 + 4J_2, \quad (305)$$

$$E_1 = C - J_1 + J_2, \quad (306)$$

$$E_2 = C + J_1 + J_2. \quad (307)$$

Write

$$H_{\text{bond}} - C = J(\cos \theta x + \sin \theta x^2), \quad J > 0. \quad (308)$$

For

$$-\frac{\pi}{4} < \theta < \frac{\pi}{4},$$

one has

$$J_1 > 0, \quad |J_2| < J_1. \quad (309)$$

The first inequality gives

$$E_2 > E_1. \quad (310)$$

The second gives

$$|2E_0 + E_2 - 3E_1| < 3(E_2 - E_1). \quad (311)$$

Its upper branch is

$$2E_0 + E_2 - 3E_1 < 3E_2 - 3E_1,$$

or

$$E_0 < E_2. \quad (312)$$

Its lower branch is

$$2E_0 + E_2 - 3E_1 > -3E_2 + 3E_1,$$

or

$$E_0 > 3E_1 - 2E_2. \quad (313)$$

Hence the complete nearest-neighbor Haldane criterion is

$$E_2 > E_1, \quad 3E_1 - 2E_2 < E_0 < E_2. \quad (314)$$

A common energy shift

$$E_J \mapsto E_J + \Lambda$$

leaves all three inequalities invariant.

At the pure $P^{(2)}$ point,

$$(E_0, E_1, E_2) = (0, 0, \Delta_2), \quad \Delta_2 > 0. \quad (315)$$

Then

$$J_1 = \frac{\Delta_2}{2}, \quad J_2 = \frac{\Delta_2}{6}, \quad (316)$$

so

$$\frac{J_2}{J_1} = \frac{1}{3}. \quad (317)$$

Therefore

$$\theta_{\text{AKLT}} = \arctan \frac{1}{3},$$

which lies strictly inside

$$-\frac{\pi}{4} < \theta < \frac{\pi}{4}.$$

Equivalently,

$$P^{(2)} = \frac{1}{6} [x^2 + 3x + 2], \quad (318)$$

so the pure maximal-spin penalty is precisely the conventional AKLT parent interaction up to an overall coefficient and additive constant [?].

F Second-Order Bosonic Hopping and the Channel Obstruction

This appendix evaluates explicitly the simplest hopping process discussed in Section 5.

Let

$$a_{ia}^\dagger, \quad a_{ja}^\dagger, \quad a = 1, 2, 3,$$

create the three components of a spin-one bosonic comparison quantum on neighboring sites i and j . Consider the spin-independent hopping Hamiltonian

$$H_t = -t \sum_{a=1}^3 (a_{ia}^\dagger a_{ja} + a_{ja}^\dagger a_{ia}). \quad (319)$$

The low-energy subspace has one boson on each site. A normalized two-site state may be written

$$|\Psi\rangle = \psi_{ab} a_{ia}^\dagger a_{jb}^\dagger |0\rangle, \quad \sum_{ab} |\psi_{ab}|^2 = 1. \quad (320)$$

Decompose

$$\psi_{ab}$$

into exchange-symmetric and exchange-antisymmetric pieces. For two spin-one representations,

$$V_1 \otimes V_1 = V_0^{\text{sym}} \oplus V_1^{\text{antisym}} \oplus V_2^{\text{sym}}. \quad (321)$$

Acting with the term that moves the boson from j to i gives

$$-t \sum_c a_{ic}^\dagger a_{jc} |\Psi\rangle = -t \psi_{ab} a_{ib}^\dagger a_{ia}^\dagger |0\rangle. \quad (322)$$

Since the creation operators commute, only the symmetric part of ψ_{ab} survives.

For a normalized symmetric internal state, define the normalized doublon state

$$|D_i[\psi]\rangle = \frac{1}{\sqrt{2}} \psi_{ab} a_{ia}^\dagger a_{ib}^\dagger |0\rangle. \quad (323)$$

Its norm is one:

$$\begin{aligned} \langle D_i[\psi] | D_i[\psi] \rangle &= \frac{1}{2} \psi_{ab}^* \psi_{cd} \langle 0 | a_{ib} a_{ia} a_{ic}^\dagger a_{id}^\dagger | 0 \rangle \\ &= \frac{1}{2} \psi_{ab}^* \psi_{cd} (\delta_{ac} \delta_{bd} + \delta_{ad} \delta_{bc}) \\ &= 1, \end{aligned} \quad (324)$$

where symmetry of ψ has been used.

Equation (322) then becomes

$$H_t |\Psi_{\text{sym}}\rangle \supset -\sqrt{2} t |D_i[\psi]\rangle. \quad (325)$$

The process moving the boson from i to j gives an independent doublon state with the same amplitude,

$$H_t |\Psi_{\text{sym}}\rangle \supset -\sqrt{2} t |D_j[\psi]\rangle. \quad (326)$$

Hence the total squared transition matrix element into the two degenerate doublon locations is

$$2 \left(\sqrt{2} t \right)^2 = 4t^2. \quad (327)$$

Let the energy cost of a same-orbital doublon in total-spin channel J be

$$U_J > 0.$$

For

$$J = 0, 2,$$

second-order perturbation theory therefore gives

$$\Delta E_J^{(2)} = -\frac{4t^2}{U_J}. \quad (328)$$

For $J = 1$, the internal wavefunction is antisymmetric:

$$\psi_{ab} = -\psi_{ba}.$$

But

$$a_{ia}^\dagger a_{ib}^\dagger$$

is symmetric in a, b , so

$$\psi_{ab} a_{ia}^\dagger a_{ib}^\dagger = 0. \quad (329)$$

Thus the same-orbital intermediate state is absent and

$$\Delta E_1^{(2)} = 0. \quad (330)$$

For the minimal hopping model,

$$\Delta E_0^{(2)} = -\frac{4t^2}{U_0}, \quad (331)$$

$$\Delta E_1^{(2)} = 0, \quad (332)$$

$$\Delta E_2^{(2)} = -\frac{4t^2}{U_2}. \quad (333)$$

If the three channels are degenerate before hopping, then for every finite positive U_2 ,

$$E_2 < E_1,$$

and therefore

$$J_1 = \frac{E_2 - E_1}{2} < 0. \quad (334)$$

The necessary Haldane condition

$$E_2 > E_1$$

is violated.

This conclusion is specific to the assumptions used above: one orbital per site, spin-independent hopping, bosonic statistics, one particle per low-energy site, and second-order exchange through same-orbital doublons. Multiple orbitals, additional direct interactions, constrained exchange, or higher-order processes can change the channel ordering.

G Symmetric-Traceless Comparison Response and the Shear Sign

Let

$$H_u$$

be the three-dimensional defect rest space. For two vector carriers, define the symmetric-traceless projector

$$(P_2)_{ab,cd} = \frac{1}{2} (h_{ac}h_{bd} + h_{ad}h_{bc}) - \frac{1}{3}h_{ab}h_{cd}. \quad (335)$$

It satisfies

$$(P_2)_{ab,cd} = (P_2)_{cd,ab}, \quad (336)$$

$$(P_2)_{ab,cd}h^{cd} = 0, \quad (337)$$

and

$$P_2^2 = P_2. \quad (338)$$

Its trace is

$$(P_2)_{ab}{}^{ab} = 5, \quad (339)$$

so its image is the five-dimensional irreducible representation

$$\text{Sym}_0^2(H_u).$$

For neighboring vector modes q_i^a, q_{i+1}^b , define the STF bond coordinate

$$Q_{ab} = (P_2)_{ab,cd}q_i^c q_{i+1}^d. \quad (340)$$

An isotropic direct comparison stiffness has the form

$$V_{\text{cmp}}^{(2)} = \frac{\Delta_{\text{cmp}}}{2} Q_{ab} Q^{ab}. \quad (341)$$

A stable positive direct penalty requires

$$\Delta_{\text{cmp}} > 0.$$

Now let

$$s_{ab} \in \text{Sym}_0^2(H_u)$$

be a geometric metric-shear mode. The most general isotropic local Gaussian mixing of the two STF sectors is

$$V^{(2)} = \frac{\Delta_{\text{cmp}}}{2} Q_{ab} Q^{ab} + \frac{K_h}{2} s_{ab} s^{ab} + \lambda s_{ab} Q^{ab}. \quad (342)$$

Assume

$$K_h > 0.$$

Completing the square gives

$$\begin{aligned} V^{(2)} &= \frac{K_h}{2} \left(s_{ab} + \frac{\lambda}{K_h} Q_{ab} \right) \left(s^{ab} + \frac{\lambda}{K_h} Q^{ab} \right) \\ &\quad + \frac{1}{2} \left(\Delta_{\text{cmp}} - \frac{\lambda^2}{K_h} \right) Q_{ab} Q^{ab}. \end{aligned} \quad (343)$$

Thus Gaussian elimination of the metric shear yields

$$\Delta_{\text{eff}} = \Delta_{\text{cmp}} - \frac{\lambda^2}{K_h}. \quad (344)$$

The induced correction is always nonpositive:

$$\delta\Delta = -\frac{\lambda^2}{K_h} \leq 0. \quad (345)$$

Therefore a positive geometric shear stiffness cannot by itself explain a positive comparison $J = 2$ penalty.

The full two-field quadratic form is positive definite precisely when

$$K_h > 0, \quad \Delta_{\text{cmp}} > 0, \quad K_h \Delta_{\text{cmp}} - \lambda^2 > 0. \quad (346)$$

Equivalently,

$$\Delta_{\text{eff}} > 0. \quad (347)$$

Hence a viable comparison-shear mechanism requires a sufficiently strong positive bare or renormalized comparison stiffness to overcome all negative induced contributions.

More generally, if the renormalized four-point comparison kernel is

$$\Gamma^{(4)} = g_0 P^{(0)} + g_1 P^{(1)} + g_2 P^{(2)}, \quad (348)$$

a common shift may be removed by defining

$$\delta_{\text{ax}} = g_1 - g_0, \quad \Delta_{\text{shear}} = g_2 - g_0. \quad (349)$$

The Haldane inequalities become

$$\Delta_{\text{shear}} > 0, \quad \delta_{\text{ax}} < \frac{2}{3} \Delta_{\text{shear}}. \quad (350)$$

The second inequality automatically implies

$$\delta_{\text{ax}} < \Delta_{\text{shear}}.$$

The ideal AKLT limit is

$$\delta_{\text{ax}} = 0, \quad \Delta_{\text{shear}} > 0. \quad (351)$$

H Projective Rotational Invariant and the AKLT Edge

Let

$$r_x \quad \text{and} \quad r_y$$

be rotations by angle π about two orthogonal axes. In $SO(3)$,

$$r_x r_y = r_y r_x. \quad (352)$$

Let

$$U_x, \quad U_y$$

be unitary representatives of their action in a projective quantum sector. Define

$$\Xi = U_x U_y U_x^{-1} U_y^{-1}. \quad (353)$$

Under arbitrary rephasings

$$U_x \mapsto e^{i\alpha} U_x, \quad U_y \mapsto e^{i\beta} U_y,$$

one has

$$\begin{aligned} \Xi &\mapsto e^{i\alpha} e^{i\beta} e^{-i\alpha} e^{-i\beta} \Xi \\ &= \Xi. \end{aligned} \quad (354)$$

Hence Ξ is rephasing invariant.

Choose canonical lifts to $SU(2)$:

$$\widetilde{r}_x = -i\sigma_x, \quad \widetilde{r}_y = -i\sigma_y. \quad (355)$$

Since

$$\sigma_x \sigma_y = i\sigma_z, \quad \sigma_y \sigma_x = -i\sigma_z,$$

the lifts anticommute:

$$\widetilde{r}_x \widetilde{r}_y = -\widetilde{r}_y \widetilde{r}_x. \quad (356)$$

Their group commutator is

$$\widetilde{r}_x \widetilde{r}_y \widetilde{r}_x^{-1} \widetilde{r}_y^{-1} = -I. \quad (357)$$

In the irreducible spin- j representation

$$\rho_j : SU(2) \rightarrow U(V_j),$$

the central element acts as

$$\rho_j(-I) = (-1)^{2j} I. \quad (358)$$

Therefore

$$\Xi_j = (-1)^{2j} I. \quad (359)$$

In particular,

$$j \in \mathbb{Z} \implies \Xi = +I, \quad (360)$$

$$j \in \mathbb{Z} + \frac{1}{2} \implies \Xi = -I. \quad (361)$$

The subgroup of $SO(3)$ generated by the three orthogonal π -rotations is the Klein four group

$$V_4.$$

Its inverse image under

$$SU(2) \rightarrow SO(3)$$

is the quaternion group

$$Q_8 = \{\pm I, \pm i\sigma_x, \pm i\sigma_y, \pm i\sigma_z\}. \quad (362)$$

Thus

$$1 \longrightarrow \mathbb{Z}_2 \longrightarrow Q_8 \longrightarrow V_4 \longrightarrow 1 \quad (363)$$

is the finite subgroup realization of the same projective obstruction.

We now apply this to the AKLT edge.

The physical microscopic carrier at each occupied comparison segment is

$$V_1.$$

It may be realized as

$$V_1 \simeq \text{Sym}^2 V_{1/2}. \quad (364)$$

Introduce two auxiliary spin- $\frac{1}{2}$ spaces at each site,

$$V_{1/2}^{(L)} \otimes V_{1/2}^{(R)},$$

and project them onto their symmetric component. Neighboring virtual spins are placed in the singlet

$$|\epsilon\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle). \quad (365)$$

This state is invariant under the diagonal $SU(2)$ action.

On a periodic chain every virtual spin is paired into a singlet. On an open chain, one virtual spin remains unpaired at each endpoint:

$$\mathcal{H}_{\text{edge},L} \simeq V_{1/2}, \quad \mathcal{H}_{\text{edge},R} \simeq V_{1/2}. \quad (366)$$

Consequently,

$$\Xi_{\text{edge},L} = -I, \quad \Xi_{\text{edge},R} = -I. \quad (367)$$

The combined two-edge sector has trivial total projective parity because

$$(-I) \otimes (-I) = +I \quad (368)$$

with respect to the central $\mathbb{Z}_2$ grading. Thus projective fractionalization at the two ends is compatible with a globally linear integer-spin microscopic Hilbert space.

At the exact AKLT point, an open chain without additional boundary terms has a four-dimensional ground-state manifold associated with the two free edge spin- $\frac{1}{2}$ factors. Away from the exact parent point but within the same symmetry-protected Haldane phase, finite-size coupling can split the edge multiplet exponentially in the chain length, while the projective edge class survives in the thermodynamic limit provided the protecting symmetry and bulk gap remain intact [?].

The relation between the microscopic and edge projective invariants is therefore

$$\Xi_{\text{microscopic}} = +I, \quad \Xi_{\text{edge}} = -I. \quad (369)$$

The half-integer projective representation is not present in the local microscopic carrier; it is generated by the entanglement structure of the many-body state.

Finally, $\Xi = -I$ should not be identified by itself with a fermionic statistics operator. It is a rotational projective invariant. A statistics grading requires additional information concerning localization and exchange. Only after a relativistic local field-theoretic completion satisfying the hypotheses of a spin-statistics theorem may the half-integer projective sector be identified with a fermionic field sector.